A PRACTITIONER'S FIELD GUIDE · BEGINNER TO INDUSTRY EXPERT

Investment Bank Treasury & Funding

How a bank funds itself, manages liquidity and collateral, prices its own balance sheet, and survives a crisis — built up from first principles to the desk-level frontier.

Treasury & Funding Masterclass
Edition 1.0 · 2026.10
Companion to the Treasury cinematic course

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01 · EXECUTIVE SUMMARY

How to Use This Guide & Executive Summary

Treasury is the function that keeps a bank alive. Trading desks make the headlines, but it is treasury that funds every asset, holds the liquidity that lets the firm open its doors each morning, posts the collateral that keeps derivatives trading, and prices the scarce balance sheet that every desk competes for. This guide takes you from zero to the desk-level frontier — from “what is a balance sheet” to pricing a KVA charge into a swap.

Who this is for

Anyone who wants to genuinely understand — and be able to *talk like an insider* about — the treasury and funding function of a large bank: new graduates on a treasury, markets or risk desk; technologists and product managers building treasury systems; investors and journalists trying to read a bank's funding profile; and experienced practitioners who want a single, rigorous reference. No prior finance degree is assumed. Every term is defined the first time it appears.

How the guide is built

The twelve chapters climb a deliberate ladder. Chapters 2–3 build the foundations: what treasury does, the balance sheet, and the toolkit a bank uses to fund itself. Chapters 4–6 are the core bank-treasury craft: transfer pricing, interest-rate risk, and the liquidity rules (LCR, NSFR) that now shape every decision. Chapters 7–9 are the advanced balance-sheet disciplines: collateral and margin, capital and leverage, and XVA. Chapters 10–12 turn to crisis history and mastery — detailed post-mortems of what broke in 2008, 2019, 2020, 2022 and 2023, then a full field catalogue of a century of liquidity crunches with their calculations, and finally what a best-in-class treasury looks like today. Chapter 13 is a glossary and reading list.

HOW TO READ IT

Read linearly the first time — each chapter assumes the last. Every quantitative idea is shown as a fully worked calculation with real numbers, every new term sits in a definition box, and every chapter ends with a “Best in class” box describing what the top institutions actually do. Come back later and the definition boxes, tables and worked examples make it a reference you can pull a single answer from.

The one-paragraph version

A bank borrows short (deposits, repo, commercial paper) and lends long (loans, bonds). That maturity transformation is profitable and permanently fragile: the bank can be perfectly solvent and still fail if it cannot roll its funding. Treasury manages that fragility on four fronts — funding (raising money at the lowest sustainable cost across the liability stack), liquidity (holding enough high-quality liquid assets to survive a 30-day run, per the LCR, and a one-year stress, per the NSFR), collateral and capital (posting margin, optimising scarce collat-

eral, and earning an adequate return on risk-weighted assets and the leverage ratio), and pricing (charging each desk the true cost of the liquidity, funding, capital and collateral it consumes, through FTP and XVA). Do all four well and the bank is a fortress; neglect any one and it is a candidate for the next war story in Chapter 10.

What you will be able to do by the end

- Read a bank's balance sheet and funding profile and say where the fragility sits.
- Calculate an LCR and an NSFR from a balance sheet, and explain every run-off factor.
- Build an FTP curve and price the true cost of a loan or the value of a deposit.
- Compute interest-rate risk two ways — ΔNII and ΔEVE — under the standard shocks.
- Work a variation-margin call, a SIMM initial-margin number, a CET1 ratio, an SLR and a CVA.
- Explain, in treasury terms, exactly what broke in 2008, September 2019, March 2020, the 2022 UK LDI crisis and the 2023 SVB failure — and what best practice now prevents it.

QUESTIONS THIS GUIDE ANSWERS

How does a bank actually fund itself, and in what order do its lenders take losses? Why can a profitable, solvent bank collapse in a week? How do regulators force a bank to hold liquidity, and how is it measured? What is collateral really doing in the modern market, and why do people say “collateral is the new cash”? And what separates a treasury that merely complies from one that is genuinely world-class?

02 · FOUNDATIONS

The Treasury Mandate & the Bank Balance Sheet

A bank treasury has one mandate above all others: fund the firm's assets at the lowest safe cost and make sure the firm can always pay what it owes, on time, in full. Everything else in this guide — repo, funds transfer pricing, interest-rate risk, Basel ratios — is machinery built to serve that single mandate. This chapter lays the first principles: what a balance sheet is, why banks earn money by taking a time mismatch, and why a perfectly solvent bank can still die of thirst.

What treasury is, and what it does

Every financial firm owns assets: loans, bonds, cash, trading positions. Assets cost money to acquire, and that money has to come from somewhere. The money that pays for assets is funding. Treasury is the function that raises funding, manages the firm's cash, and guarantees that every payment is met the moment it falls due.

> Treasury — the central function that raises cash to pay for the firm's assets and maturing debts, chooses the cheapest safe source of that cash, holds a buffer to survive stress, and ensures the firm never misses a payment.

Note what treasury is *not*. It is not a profit-chasing desk. Its north star is survival at minimum cost, not return maximisation. A treasurer who earns an extra five basis points by funding a 10-year mortgage book overnight has not been clever; they have planted a time bomb. The craft is to pay as little as possible for funding subject to the firm never running out of cash.

Day to day, treasury does three things. First, it raises cash to pay for new assets and to repay maturing liabilities. Second, it picks the cheapest source that is also safe and sustainable — secured before unsecured, diversified rather than concentrated, term-matched to the assets being funded. Third, it holds a liquidity buffer so the firm survives a day when markets freeze and lenders refuse to roll.

The bank balance sheet

Everything a bank does fits one identity. Learn it once; it governs the rest of the guide.

> The balance-sheet identity — Assets = Liabilities + Equity. Assets are what the firm owns; liabilities are what the firm owes; equity is the owners' residual stake, the amount left over after every creditor is paid.

A bank's assets are dominated by cash and central-bank reserves, loans to customers, and a portfolio of securities (government bonds, agency paper, corporate credit). Its liabilities are the funding that pays for those assets: customer deposits, wholesale funding such as bonds and commercial paper, and secured borrowing such as repo. The difference — assets minus liabilities — is equity, also called capital. Equity is the loss-absorbing

cushion. Losses hit equity first. Only when losses exceed equity does the bank become insolvent, meaning its liabilities exceed its assets.

WORKED EXAMPLE

A stylised mid-size bank, figures in billions.

Assets	$bn	Liabilities & equity	$bn
Cash & central-bank reserves	60	Retail deposits	420
Securities (govvies, agency)	180	Wholesale funding (bonds, CP)	240
Loans to customers	620	Repo borrowing	130
Other assets	30	Equity (CET1 + other capital)	100
Total assets	890	Total liabilities + equity	890

The two sides equal by construction. Equity of $100bn supports $890bn of assets — a leverage of roughly 8.9×. If loan losses of $40bn crystallise, equity falls from $100bn to $60bn and the identity re-balances; the bank is wounded but still solvent. If losses reached $110bn, equity would be wiped out and the bank would be insolvent, with $780bn of assets against $790bn of liabilities.

Maturity transformation: borrow short, lend long

Banks earn money from a time gap. They borrow for short terms — deposits that can leave tomorrow, repo that rolls overnight — and they lend or invest for long terms: a 25-year mortgage, a 5-year corporate loan, a 10-year bond. The long rate normally sits above the short rate, so the bank pockets the spread. This is maturity transformation.

> Maturity transformation — funding long-dated assets with short-dated liabilities, so the firm earns the gap between the long rate it receives and the short rate it pays.

The gap is profit, and the gap is also the risk. Short funding must be renewed again and again. If lenders stop renewing — a run — the bank cannot repay the short money, yet it still holds long assets it cannot sell quickly without a fire-sale loss. This is funding risk, and it is the risk a treasury exists to contain. The fat interest margin and the risk of sudden death are two faces of the same mismatch; never read one without the other.

Net interest margin

The profitability of maturity transformation is captured in one headline number.

> Net interest margin (NIM) — net interest income (interest earned on assets minus interest paid on funding) divided by average interest-earning assets, expressed as a percentage. It measures how many cents of net interest the bank earns per dollar of earning assets.

WORKED EXAMPLE

Take the stylised bank's earning assets of $800bn (the $890bn total less $30bn of non-earning other assets and $60bn held as low-yielding cash buffer — assume $800bn earns a blended yield).

```
Interest income   = $800bn × 4.60%  = $36.80bn
Interest expense  = $790bn × 2.85%  = $22.52bn
Net interest income (NII)           = $14.28bn

NIM = NII / average earning assets
    = $14.28bn / $800bn
    = 1.785%  ≈ 1.79%
```

The bank earns a net 179 basis points on every dollar of earning assets. A one-basis-point move in the blended funding cost on $790bn of liabilities is $79m of pre-tax profit — which is exactly why treasury obsesses over the funding curve and why FTP (Chapter 04) pushes funding cost down to every desk.

Liquidity versus solvency — why a solvent bank can still fail

These two words are confused constantly, and the confusion is fatal. Keep them rigidly separate.

> Solvency — a balance-sheet condition: assets exceed liabilities, so equity is positive. A solvent firm, given time, could pay everyone.

> Liquidity — a cash-timing condition: the ability to produce cash to meet obligations as they fall due. A liquid firm can pay *today*.

These are different problems with different cures. A firm can be deeply solvent and still fail because it cannot find cash in the hour it is needed. Most bank failures are liquidity deaths, not solvency deaths — the 2023 failures of Silicon Valley Bank and others were triggered by deposits fleeing faster than assets could be converted to cash, not by the book being underwater on day one.

WORKED EXAMPLE

A solvent bank runs out of cash. Start from the stylised balance sheet: $890bn assets, $790bn liabilities, $100bn equity — clearly solvent. Now a confidence shock hits and over five days $150bn of deposits and wholesale funding flee.

```
Immediately usable cash        = $60bn reserves
Securities sellable in 5 days  = $180bn, but fire-sale
                                 at a 7% stress discount
                               → $180bn × 0.93 = $167.4bn
Total cash raisable fast       = $60bn + $167.4bn = $227.4bn

Outflow over 5 days            = $150bn
Cash shortfall on day 1–2      = outflows arrive faster than
                                 the $167.4bn of sales settle
```

On paper the bank can cover $150bn of outflow from $227.4bn of raisable cash. But selling $150bn of bonds in two days is impossible without crushing the price, and much of the securities book may already be encumbered (pledged in repo), so it is not free to sell. The bank is solvent — assets still exceed liabilities even after the fire-sale haircut — yet it misses a settlement and defaults. Solvency did not save it; liquidity killed it. This single scenario is the reason the LCR (Chapter 06) exists.

The three treasury sub-functions

A treasury is really three disciplines working in concert.

- Funding. Raising cash in the money markets every morning — deposits, repo, commercial paper, senior unsecured — at the lowest safe blended cost. The subject of Chapter 03.
- ALM & liquidity. Managing the interest-rate and liquidity mismatches across the whole balance sheet: repricing gaps, EVE/NII sensitivity, the HQLA buffer, LCR and NSFR. Chapters 05 and 06.
- Collateral & capital. Mobilising collateral for secured funding and margin, tracking encumbrance, and respecting the capital and leverage constraints (CET1, SLR) that gate how large the balance sheet can grow.

The stakeholders

Treasury does not act alone. It answers to and negotiates with a defined set of stakeholders.

> ALCO (Asset-Liability Committee) — the senior committee that governs ALM. It meets monthly or more often, sets risk limits, approves the funding plan, and reviews liquidity and interest-rate reports. Members typically include the CFO, the Treasurer, the Chief Risk Officer, and the heads of the major business lines.

- Trading and lending desks consume funding and provide deposits; treasury prices them internally through FTP so each desk sees the true cost of the liquidity it uses.
- Regulators (the Fed, PRA, ECB, and national supervisors) set and monitor the binding ratios — LCR, NSFR, leverage, capital — and run the stress tests.
- Depositors and wholesale lenders supply the funding and can withdraw it; their confidence is the bank's lifeblood.
- Rating agencies and the market price the bank's unsecured spread, which feeds straight back into the cost of funds.

BEST IN CLASS

The strongest treasuries run two clocks simultaneously and never let either stop. The earnings clock keeps NII and EVE inside tight limits as rates move; the cash clock keeps enough *unencumbered* HQLA to survive a 30-day run, fund structurally for a year, and bridge every intraday payment peak. They diversify funding across retail, wholesale, secured and unsecured channels and across tenors so no single source can break them. They treat the balance sheet as a scarce, priced resource — every desk is charged for the liquidity and capital it consumes — and they maintain a tested contingency funding plan so that when confidence breaks, the playbook is already written. Above all, the best never confuse a healthy capital ratio with cash in the till: they know a solvent bank dies of a liquidity drought, so they measure usable liquidity, not headline buffers.

03 · FUNDING TOOLKIT

The Money Markets & the Liability Stack

The money markets are where a treasury raises short-term cash every morning — anything from overnight to about one year. This chapter walks the full toolkit: unsecured funding on the firm's credit, secured funding against collateral, the overnight rate stack that prices it all, FX swaps as synthetic funding, and the loss-absorbing waterfall that ranks who takes the hit when a bank fails.

Unsecured funding — borrowing on the firm's promise

> Unsecured funding — cash lent on the borrower's promise to repay, with no collateral posted. The lender relies entirely on the borrower's creditworthiness and ranks as a general creditor if the borrower defaults.

Unsecured funding is the most expensive and most fragile part of the stack because the lender bears the bank's credit risk directly. It comes in several forms:

- Retail deposits — balances from individuals and small businesses. Small per account, but sticky: insured up to the deposit-guarantee limit and tied to a banking relationship, so they rarely all leave at once. Regulators treat them as the most reliable funding (lowest run-off in the LCR).
- Wholesale deposits — large balances from banks, funds, and corporates. Big but flighty; managed by professional treasurers who move at the first sign of stress.
- Interbank deposits — one bank places cash with another, overnight or for a term, priced off the borrowing bank's credit.
- Commercial paper (CP) — short-term unsecured notes, typically 1–90 days, issued at a discount and redeemed at par.
- Certificates of deposit (CDs) — tradable deposit receipts for a fixed term at a fixed rate, sellable in the secondary market before maturity.
- Senior unsecured bonds — longer-dated debt that forms the backbone of term funding and the senior rung of the loss waterfall.

WORKED EXAMPLE

Commercial-paper discount yield. A bank issues 90-day CP with a $10,000,000 face (par) value and sells it today for $9,880,000. CP is quoted on a discount basis, ACT/360.

```
Discount           = Face - Price
                   = $10,000,000 - $9,880,000 = $120,000

Discount rate      = (Discount / Face) × (360 / days)
                   = ($120,000 / $10,000,000) × (360 / 90)
                   = 0.0120 × 4 = 4.80%

True money-market yield (on price invested):
Yield              = (Discount / Price) × (360 / days)
                   = ($120,000 / $9,880,000) × 4
                   = 0.012146 × 4 = 4.86%
```

The quoted discount rate is 4.80%, but the investor's actual return on the $9.88m paid is 4.86% — the discount convention always understates the true yield, because the discount is measured against face, not against the smaller amount actually invested.

Secured funding — borrowing against collateral

Most wholesale funding today is secured: the borrower pledges collateral the lender can keep and sell on default. Because the lender's risk is lower, secured funding is cheaper. The dominant instrument is the repo.

> Repurchase agreement (repo) — a sale of securities with an agreement to buy them back later at a higher price. Economically it is a collateralised loan: the seller borrows cash and posts securities; the buyer lends cash and holds the collateral. The price difference is the interest, called the repo rate. The same trade seen from the cash lender's side is a reverse repo.

> Haircut (initial margin) — the lender advances less cash than the collateral is worth, so a 2% haircut on a $10m bond means $9.8m of cash. The buffer protects the lender if the collateral's price falls before it can be liquidated.

WORKED EXAMPLE

A 7-day repo on a $10m government bond. Repo rate 5.00%, ACT/360, haircut 2%.

```
Step 1 — cash advanced (apply haircut)
  $10,000,000 × (1 - 0.02) = $9,800,000

Step 2 — repo interest (ACT/360, 7 days)
  $9,800,000 × 5.00% × (7 / 360) = $9,527.78

Step 3 — cash repaid at maturity
  $9,800,000 + $9,527.78 = $9,809,527.78
```

The borrower receives $9.8m on day 0 and repays $9,809,527.78 on day 7 to retrieve the bond. The $200,000 haircut stays over-collateralising the lender throughout — which is why a firm that funds entirely in repo ties up more collateral than the cash it raises, and why rollover risk bites if lenders refuse to re-lend.

> General Collateral (GC) vs special — under GC the lender accepts any bond from a broad agreed basket and just wants cash safely invested, so GC trades near the general secured rate. A special is a single security in high demand to borrow; cash lenders accept a *lower* repo rate to get hold of it, so specials trade below GC — occasionally at negative repo rates when the shortage is extreme.

The overnight rate stack

Every funding trade is priced off a reference rate. The modern short-end stack, from secured floor to unsecured top, is the mental model every treasurer carries.

> SOFR (Secured Overnight Financing Rate) — the volume-weighted median of overnight Treasury repo across tri-party, GCF, and FICC-cleared bilateral — roughly $2 trillion of daily volume. It is secured and near risk-free, which is why it replaced LIBOR and why, by design, it says nothing about *unsecured* bank funding stress.

> Fed funds — the rate on unsecured overnight lending of reserves between banks; the Fed targets a range for it. OIS (overnight index swap) — a swap whose floating leg compounds a daily overnight rate (SOFR or fed funds); the fixed rate on an OIS is the market's expected average overnight rate over the term, and is the cleanest proxy for the risk-free curve.

- ON RRP (the Fed's reverse-repo facility) sits at the *floor* — a reservation rate for money funds.
- Tri-party GC and SOFR print just above the floor in normal times.
- IORB (interest on reserve balances) sits near the top as an administered ceiling-ish rate.

- Specials and basis-trade financing can reach above GC in the bilateral market.

The gap between an unsecured bank-funding rate and the secured/risk-free rate is the secured–unsecured basis. Watched as FRA-OIS (unsecured bank funding stress) or the Treasury-OIS spread, it widens violently in a crisis as unsecured lenders demand a premium for bank credit. When the whole repo stack lifts off the RRP floor, it signals that bank reserves are becoming scarce.

FX swaps as synthetic funding

> FX swap — a two-legged trade that funds one currency using another: at the start you exchange currency A for currency B at spot; at maturity you reverse at an agreed forward rate. Between the legs you hold the currency you needed. A euro-rich bank can thereby raise dollars without issuing dollar debt.

In theory, borrowing dollars synthetically through an FX swap should cost exactly the same as borrowing dollars directly — this is covered interest parity (CIP). In practice CIP breaks. When many firms scramble for dollars, the synthetic route costs *more* than the interest-rate differential implies, and the cross-currency basis turns negative. The BIS has flagged the enormous stock of off-balance-sheet FX-swap funding as a hidden dollar-funding risk. Treasury watches the basis daily to pick the cheaper of direct versus synthetic dollars.

WORKED EXAMPLE

Implied USD funding cost via an EUR/USD FX swap (CIP). A bank holds euros and needs 1-year dollars.

```
Inputs
  Spot EUR/USD            = 1.0800
  1y EUR interest rate    = 3.40%
  1y forward EUR/USD      = 1.0640  (dollar at a forward premium)

CIP forward (no-basis):  F = S × (1 + r_USD) / (1 + r_EUR)
Solve for the implied USD rate the swap actually charges:
  1 + r_USD(implied) = (F / S) × (1 + r_EUR)
                     = (1.0640 / 1.0800) × 1.0340
                     = 0.98519 × 1.0340 = 1.01868
  r_USD(implied)     = 1.868%  ≈ 1.87%
```

If direct 1-year USD funding (e.g. SOFR-OIS based) is only 1.55%, the swap is charging 1.87% — about 32bp of negative cross-currency basis, the extra cost of obtaining dollars synthetically. The treasurer funds directly in dollars and avoids paying the basis.

The liability / loss-absorbing waterfall

When a bank fails, losses do not hit creditors at random; they climb a defined ladder from the bottom up. Equity absorbs first; insured depositors are protected last. This ordering is what gives secured and senior funding its low cost and what makes AT1 and equity expensive.

> TLAC / MREL — Total Loss-Absorbing Capacity (global) and the Minimum Requirement for own funds and Eligible Liabilities (EU): a mandated layer of capital plus bail-in-able debt that can be written down or converted to recapitalise a failing bank in resolution, protecting depositors and taxpayers.

> AT1 / CoCo — Additional Tier 1, or contingent convertible bonds: perpetual instruments that convert to equity or are written down if the bank's CET1 ratio breaches a trigger. They sit just above equity and absorb losses while the bank is still a going concern.

Rung (loss order)	Instrument	Absorbs loss
1 — first loss	CET1 equity	Immediately, going concern
2	AT1 / CoCo	At CET1 trigger
3	Tier 2 subordinated	In resolution
4	TLAC / MREL senior non-preferred	Bail-in in resolution
5	Senior unsecured	After bail-in layers
6 — last loss	Insured deposits (& secured)	Protected / collateralised

WORKED EXAMPLE

Where a $70bn loss lands. A bank has, from the bottom of the waterfall: CET1 equity $45bn, AT1 $12bn, Tier 2 $8bn, TLAC senior non-preferred $25bn.

```
Loss to absorb                          = $70bn
  1. CET1 equity  $45bn  → fully absorbed, $45bn used
     Remaining loss                     = $70 - $45 = $25bn
  2. AT1 $12bn    → written down in full, $12bn used
     Remaining loss                     = $25 - $12 = $13bn
  3. Tier 2 $8bn  → written down in full, $8bn used
     Remaining loss                     = $13 - $8 = $5bn
  4. TLAC SNP     → $5bn bailed in (of $25bn available)
```

Equity, AT1 and Tier 2 are entirely wiped; $5bn of the TLAC/MREL layer is bailed in. Senior preferred bondholders and all insured depositors are untouched — exactly the design intent. This waterfall is why insured deposits fund cheapest and AT1 pays the fattest coupon.

BEST IN CLASS

Elite funding desks never treat the money market as a single tap. They ladder maturities so no cliff of rollovers ever lands on one day, and they keep deliberate diversity across retail, operational wholesale, secured repo, FX-swap and term-bond channels so a freeze in any one venue is survivable. They price every funding route against the live curve each morning — GC repo versus tri-party versus cross-currency basis — and route to the cheapest safe source rather than the habitual one. They manage collateral as a scarce asset, keeping the best paper unencumbered for the contingency plan, and they issue term debt and TLAC/MREL ahead of need, not in the teeth of a crisis when spreads have already blown out. The signature of mastery is a funding profile that is boring on a good day and still boring on a bad one.

04 · FTP

Funds Transfer Pricing

Funds Transfer Pricing is the internal plumbing that makes every business line see the true cost — or true value — of the liquidity it consumes or provides. Without it, a lending desk can look wildly profitable while quietly living off cheap central funding, and a deposit-gathering branch can look worthless while supplying the firm's most valuable resource. FTP fixes the incentives by charging and crediting every desk at a transfer rate drawn from the firm's own funding curve.

What FTP is and why it exists

> Funds Transfer Pricing (FTP) — the mechanism by which a bank's central treasury charges asset-generating desks for the funds they use and credits liability-generating desks for the funds they provide, both at a transfer rate taken from the firm's funding curve and matched to the tenor of the position.

The architecture is simple: treasury runs a single central funding book. No desk touches the market directly. Instead, every desk transacts with treasury. A lending desk that writes a 5-year loan "borrows" the cash from treasury and is charged the matched transfer rate. A deposit desk that gathers a 3-month deposit "lends" that cash to treasury and is credited the matched transfer rate. Treasury then funds only the *net* gap in the external market and owns the residual interest-rate and liquidity risk.

FTP does three jobs. It strips funding guesswork out of every desk's P&L. It prices liquidity and tenor consistently across the whole firm, so no desk games an internally subsidised rate. And it reveals the true profitability of each business after funding is paid for.

The matched-maturity method

> Matched-maturity FTP — each cash flow is priced at the point on the funding curve that matches its own maturity (or repricing tenor), so a 5-year fixed asset is charged the 5-year funding rate and a 3-month deposit is credited the 3-month rate. This isolates each desk's business margin from treasury's maturity-transformation result.

The principle is that a desk should be neither rewarded nor penalised for the shape of the yield curve — that mismatch belongs to treasury, which is paid to run it. Matched-maturity pricing hands each desk a clean, curve-neutral margin and parks the entire term-transformation gain or loss in the treasury book where it can be hedged and governed by ALCO.

The components of the transfer rate

A full FTP charge on an asset is not just the base rate. It is built in layers.

- Base / matched-maturity funding rate — the risk-free or OIS curve point at the asset's tenor.
- Liquidity (term) premium — an add-on reflecting the extra cost of raising *term* funding rather than overnight money; longer, less-liquid assets pay more.
- Contingent-liquidity charge — a fee for the standby liquidity the firm must hold against undrawn commitments and other optionality the desk has sold.

> Liquidity (term) premium — the incremental spread over the risk-free curve that the firm pays to lock in funding for a longer term. It compensates the funder for tying up money and protects the firm against rollover risk; it is the price of the maturity match itself.

WORKED EXAMPLE

Full FTP charge build-up for a 5-year fixed-rate loan funded against the curve.

```
Component                                  Rate
  5y matched-maturity base (OIS/swap curve)   3.60%
  + Term liquidity premium (5y)             + 0.45%
  + Contingent-liquidity / buffer cost      + 0.10%
  ------------------------------------------------
  All-in FTP charge to the lending desk     = 4.15%

Loan is booked to the customer at           5.95%
  - All-in FTP charge                       - 4.15%
  ------------------------------------------------
  Net lending (business) margin             = 1.80%
```

The lending desk keeps a clean 1.80% margin that is independent of what rates do, because treasury has already charged it the matched 5-year cost plus the liquidity and buffer add-ons. The desk is rewarded only for credit selection and pricing power — exactly the behaviour the bank wants.

Behavioural maturity of non-maturity deposits

> Non-maturity deposit (NMD) — a deposit with no contractual end date, such as a checking or savings account; the customer may withdraw any day, yet in practice most balances persist for years. FTP must assign a *behavioural* maturity to decide which point of the curve credits the deposit.

This is where FTP becomes an art. A checking balance is contractually overnight, but if treasury credited it only the overnight rate it would massively understate its value — the stable core behaves like multi-year funding. So

treasury behaviouralises the pool: it splits balances into a stable core, credited at a longer tenor, and a volatile layer, credited short. Get this wrong in either direction and the firm misprices its single largest funding source.

WORKED EXAMPLE

The FTP credit a 3-month wholesale deposit earns, versus a behaviouralised savings pool.

```
Case A — contractual 3-month deposit of $500m
  3m matched funding-curve credit rate        3.15%
  FTP credit to the deposit desk
    = $500m × 3.15% × (90 / 360)             = $3.9375m per quarter

Case B — $500m savings pool, behaviouralised
  Stable core 70% → credited at 3y rate         3.40%
  Volatile 30%    → credited at 1m rate         3.05%
  Blended FTP credit rate
    = 0.70 × 3.40% + 0.30 × 3.05%            = 3.295%
  Annual FTP credit = $500m × 3.295%         = $16.475m/yr
```

The behaviouralised savings pool earns the desk a blended 3.295% — materially above the pure overnight rate — because treasury recognises that 70% of it is, in practice, term funding. This credit is what makes retail deposit-gathering a valued business inside the firm.

Contingent-liquidity charges for undrawn commitments

A committed-but-undrawn credit line is a liquidity liability even though no cash has moved: the client can draw at the worst possible moment, precisely when the firm is itself under stress. Treasury must hold HQLA against that contingent draw (the LCR assigns undrawn committed financial facilities a 40% draw factor). FTP charges the originating desk a contingent-liquidity fee for the standby buffer, so the desk internalises the cost of the optionality it has sold rather than treating undrawn lines as free.

How FTP steers the balance sheet

Because the transfer rate is a price, treasury can use it as a steering wheel. Raise the term-liquidity premium and long-dated, illiquid lending becomes less attractive at the margin; raise the deposit credit and the branches push harder for sticky retail balances. FTP is therefore not just an accounting allocation — it is the firm's internal monetary policy, aligning thousands of individual desk decisions with the treasurer's view of where the balance sheet should grow and shrink.

WORKED EXAMPLE

Decomposing a loan's NIM into its three economic parts. A 5-year loan yields 5.95% to the customer; the desk's matched FTP charge is 4.15%; the firm's blended external funding cost for that 5-year money is 3.95% (it funds a touch cheaper than the internal transfer rate because of its deposit franchise).

```
Customer loan rate                          5.95%
  - Lending spread (credit + origination)   kept by lending desk
  - Treasury/liquidity spread               kept by treasury
  - External funding cost                   paid to the market

Lending (business) margin  = 5.95% - 4.15%   = 1.80%  → lending desk
Treasury/liquidity spread  = 4.15% - 3.95%   = 0.20%  → treasury book
External funding cost      = 3.95%                     → the market
Check: 1.80% + 0.20% + 3.95%                 = 5.95%  ✓
```

The 5.95% NIM contribution decomposes cleanly: 1.80% is the lending desk's business margin, 0.20% is treasury's liquidity/transformation spread, and 3.95% is the raw cost of money. Each owner is now accountable for exactly the slice they control.

BEST IN CLASS

Leading banks run FTP as a single, fully matched-maturity framework with no exceptions and no hand-negotiated rates — the curve, the term-liquidity premium, and the contingent-liquidity charge are published and applied mechanically, which is what gives the numbers their authority in a P&L dispute. They behaviouralise non-maturity deposits with models that are back-tested against real decay and re-calibrated after every rate cycle, and they charge undrawn commitments the true cost of the standby buffer so no free options accumulate on the book. Critically, they treat FTP as a steering instrument owned by ALCO, not a back-office allocation: when the treasurer wants to lengthen funding or slow illiquid lending, the lever is the transfer rate. The tell of a weak shop is a flat, single-rate FTP that quietly subsidises the longest, least-liquid assets — the very assets that sink banks.

05 · ALM & IRRBB

Asset-Liability Management & Interest-Rate Risk in the Banking Book

A bank that borrows short and lends long is, by construction, exposed to interest rates: its assets and liabilities reprice at different speeds. ALM is the discipline of measuring and shaping that exposure so a rate move does not wreck either near-term earnings or the long-run value of the firm. This chapter covers the repricing gap, duration, the EVE and NII lenses, the standard regulatory shocks, deposit beta, embedded optionality, and hedging with swaps.

The two lenses: NII and EVE

> IRRBB (Interest-Rate Risk in the Banking Book) — the risk that changes in interest rates damage a bank through its banking-book assets and liabilities, which reprice on mismatched schedules. It is managed through two complementary measures: NII and EVE.

> Net Interest Income (NII) sensitivity — the earnings lens: how much net interest income changes over the next one-to-two years for a given rate move. Short-horizon, cash-flow focused.

> Economic Value of Equity (EVE) — the value lens: the present value of all banking-book asset cash flows minus the present value of all liability cash flows. EVE sensitivity measures how that net present value moves when rates shift. Long-horizon, full-term-structure focused.

A bank can look safe on one lens and dangerously exposed on the other, so treasury watches both. A long fixed-rate bond book funded by overnight deposits has a large negative EVE sensitivity to rising rates (long-duration assets lose value) even if near-term NII looks fine because deposits are slow to reprice.

The repricing gap

> Repricing gap — assets and liabilities are sorted into time buckets by when they next reprice; the gap in each bucket is rate-sensitive assets minus rate-sensitive liabilities. A positive (asset-sensitive) gap gains NII when rates rise; a negative (liability-sensitive) gap loses NII when rates rise.

WORKED EXAMPLE

A repricing-gap table and the ΔNII from a +200bp shock. Figures in $bn; the ΔNII column applies the shock to the cumulative gap, weighted by the fraction of the year the bucket is exposed.

Bucket	RSA	RSL	Gap	Yr fraction	ΔNII @ +200bp
0–3m	120	180	−60	0.875	−1.050
3–6m	90	110	−20	0.625	−0.250
6–12m	140	95	+45	0.250	+0.225
≤1y total	350	385	−35	—	−1.075

```
ΔNII (bucket) = Gap × shock × year-fraction remaining
  0–3m : -60 × 0.0200 × 0.875 = -$1.050bn
  3–6m : -20 × 0.0200 × 0.625 = -$0.250bn
  6–12m: +45 × 0.0200 × 0.250 = +$0.225bn
  Total 12-month ΔNII          = -$1.075bn
```

The bank is liability-sensitive inside a year (net −$35bn gap), so a +200bp shock cuts net interest income by about $1.075bn over the next twelve months. Liabilities reprice up faster than assets. Treasury would hedge this by receiving fixed / paying floating, or by lengthening liability duration.

Duration, modified duration, and EVE

> Duration — the weighted-average time to receive a bond's cash flows, in years; it measures how long capital is tied up. Modified duration converts that into price sensitivity: the approximate percentage change in value for a 1% change in yield, equal to duration ÷ (1 + yield).

> Convexity — the curvature in the price-yield relationship that duration alone misses; it makes the duration estimate slightly understate gains when rates fall and overstate losses when rates rise. It matters most for instruments with embedded options.

WORKED EXAMPLE

EVE sensitivity from duration. A bank's banking book:

```
Assets      = $800bn, modified duration 4.5 years
Liabilities = $720bn, modified duration 1.8 years
Equity (economic) = $80bn

Shock = +200bp = +0.0200

ΔValue(assets)      = -D_A × Δy × Assets
                    = -4.5 × 0.0200 × $800bn = -$72.0bn
ΔValue(liabilities) = -D_L × Δy × Liabilities
                    = -1.8 × 0.0200 × $720bn = -$25.92bn
                      (liabilities lose value too, which HELPS equity)

ΔEVE = ΔValue(assets) - ΔValue(liabilities)
     = (-$72.0bn) - (-$25.92bn)
     = -$46.08bn
```

A +200bp shock erases roughly $46.1bn of economic value of equity — about 58% of the $80bn economic equity, far beyond the regulatory ±15%-of-Tier-1 outlier threshold. The culprit is the duration mismatch: assets at 4.5 years versus liabilities at 1.8 years. The fix is to shorten asset duration or lengthen liability duration, typically with swaps.

The standard rate shocks

Regulators do not rely on a single move. The Basel IRRBB framework specifies six standardised shocks applied to the yield curve: parallel up, parallel down, steepener, flattener, short-rate up, and short-rate down. The headline ±200bp parallel shock is the workhorse, but the non-parallel shocks catch banks that are hedged against a parallel move yet exposed to a change in curve shape. EVE sensitivity across all six is compared against an outlier test of ±15% of Tier 1 capital; a breach invites supervisory action.

Deposit beta and non-maturity-deposit modelling

> Deposit beta — the fraction of a market rate move that a bank passes through to its deposit rates. A beta of 0.50 means a +200bp rise in market rates lifts deposit rates by only +100bp; the other 100bp is retained as margin. Low beta is good for NII but tends to erode as competition and customer attention rise.

Because non-maturity deposits have no contractual repricing date, their beta is an *assumption*, and it is the single most contested input in the whole IRRBB model. Underestimate beta and the bank overstates how much of a rate rise it keeps; the March 2023 episode showed how fast depositors can demand higher rates or leave entirely once they pay attention.

WORKED EXAMPLE

The effect of a 50% vs 80% deposit beta on NII. The bank holds $400bn of non-maturity deposits; market rates rise +200bp. Assets reprice fully (+200bp on $350bn of rate-sensitive assets, for simplicity assume earning yield on the deposit-funded book rises by the full +200bp on $400bn of matched assets).

```
Asset income uplift = $400bn × +2.00%            = +$8.00bn

Deposit cost uplift depends on beta:
  Beta 0.50 : $400bn × (2.00% × 0.50) = $400bn × 1.00% = +$4.00bn
  Beta 0.80 : $400bn × (2.00% × 0.80) = $400bn × 1.60% = +$6.40bn

Net interest income change:
  Beta 0.50 : +$8.00bn - $4.00bn = +$4.00bn
  Beta 0.80 : +$8.00bn - $6.40bn = +$1.60bn

Difference                                        = $2.40bn
```

Moving the assumed beta from 50% to 80% cuts the NII benefit of a +200bp rise from $4.0bn to $1.6bn — a $2.4bn swing driven purely by a behavioural assumption. This is why supervisors scrutinise deposit-beta modelling so hard: the whole IRRBB result hinges on it.

Embedded optionality

Banking-book cash flows are rarely fixed. Mortgages carry a prepayment option — borrowers refinance when rates fall, shortening asset duration exactly when the bank would prefer it long. Loans and deposits carry caps and floors that truncate how far rates pass through. These options create the convexity that a simple gap or duration measure misses, and they almost always work against the bank (customers exercise in their own favour). Treasury models prepayment speeds and option-adjusted duration precisely because ignoring the optionality systematically understates risk.

Hedging with interest-rate swaps

> Interest-rate swap (IRS) — an agreement to exchange a stream of fixed-rate payments for a stream of floating-rate payments (today referenced to SOFR) on a notional amount. It lets a bank change the effective duration of its book without buying or selling the underlying assets.

A liability-sensitive bank — one that loses NII when rates rise — hedges by entering a swap to receive fixed and pay floating, which synthetically lengthens its liability duration and offsets the gap. A bank with too much fixed-rate asset duration does the reverse: pay fixed, receive floating, to shorten effective asset duration. The swap moves the EVE and NII sensitivities toward the ALCO-approved limits while leaving the customer-facing balance sheet untouched. Hedging is cheap and surgical compared with restructuring the loan and deposit book, which is why swaps are the primary IRRBB tool.

BEST IN CLASS

The strongest ALM functions never trust a single metric: they run NII and EVE together, across all six standardised curve shocks plus their own larger internal scenarios, and they reconcile the two lenses rather than cherry-picking the flattering one. They treat deposit beta and non-maturity-deposit decay as living, back-tested models re-calibrated after every rate cycle — and they stress the *assumptions*, not just the rates, because that is where the real risk hides. They explicitly model prepayment and cap/floor optionality so convexity never ambushes them. And they hedge surgically with swaps to hold EVE and NII inside tight ALCO limits, keeping the maturity-transformation profit while refusing to bet the firm on a single rate path. The weak shop is the one that reported a comfortable parallel-shock number in 2021 and discovered in 2023 that its deposits were neither as cheap nor as sticky as the model assumed.

06 · LIQUIDITY REGULATION

Liquidity Regulation — LCR & NSFR

After 2008, Basel III added two binding liquidity standards to sit alongside the capital rules. The Liquidity Coverage Ratio asks whether a bank can survive a 30-day cash sprint; the Net Stable Funding Ratio asks whether its funding structure can survive a one-year marathon. Together they force banks to hold a buffer of high-quality liquid assets and to fund illiquid assets with stable money. This chapter works both ratios end to end.

HQLA — the numerator of the LCR

> HQLA (High-Quality Liquid Assets) — assets that stay liquid and hold value in stress, convertible to cash with little or no loss. Basel III sorts them into tiers with rising haircuts and tightening caps; only unencumbered HQLA counts.

Tier	Examples	Haircut	Cap
Level 1	Cash, central-bank reserves, high-grade sovereigns	0%	No cap
Level 2A	High-grade agency/sovereign, qualifying covered bonds	15%	L2 ≤ 40%
Level 2B	Lower-rated corporates, qualifying equities	25–50%	L2B ≤ 15%

Two caps bite. Total Level 2 may not exceed 40% of the buffer, and Level 2B may not exceed 15%. These force the buffer to lean on the safest paper and stop a bank from dressing up a thin Level 1 base with volatile equities.

Net cash outflows — the denominator

> Run-off factor — the share of a funding source assumed to flee within the 30-day stress window. Sticky money runs slowly; hot money runs fast. The denominator is a *modelled* stress, not actual flows.

Funding / exposure	30-day run-off (outflow)
Stable retail deposits (insured, operational)	3–5%
Less-stable retail deposits	10%+
Operational wholesale (clearing, custody)	25%
Non-operational corporate deposits	40%
Unsecured funding from financials	100%
Secured funding against Level 1 (vs market)	0%
Secured funding against non-HQLA	100%
Undrawn committed facilities (financial)	40%

Inflows (e.g. maturing loans the bank expects to be repaid) offset outflows, but with a hard cap: inflows may offset no more than 75% of outflows. This guarantees a bank always holds real HQLA and can never run purely on expected receipts.

> Liquidity Coverage Ratio (LCR) — HQLA (after haircuts) ÷ net cash outflows over a 30-day stress ≥ 100%. It tests whether the bank can survive one month of acute stress on its own standalone liquidity.

WORKED EXAMPLE

A full LCR calculation from a sample balance sheet. Figures in $bn.

```
NUMERATOR — HQLA after haircuts and caps
  Level 1  : cash+reserves+sovereigns  $80 × 100% = $80.0
  Level 2A : agency/covered   $20 × (1 - 0.15)     = $17.0
  Level 2B : qualifying equities $6 × (1 - 0.50)   = $3.0
  Cap check: L2 = 17 + 3 = $20 vs 40% of total buffer
             40% cap on ($80+$20) → L2 ≤ $66.7  OK
             L2B cap 15% → $3 ≤ $15  OK
  Total HQLA                                        = $100.0

DENOMINATOR — net 30-day stressed outflows
  Stable retail     $200 × 5%                 = $10.0
  Less-stable retail $50 × 10%                = $5.0
  Operational whsl  $60 × 25%                 = $15.0
  Non-op corporate  $100 × 40%                = $40.0
  Financial unsec.  $30 × 100%                = $30.0
  Undrawn lines     $25 × 40%                 = $10.0
  Gross outflows                              = $110.0

  Expected inflows                            = $28.0
  Inflow cap = 75% × $110 = $82.5 → use $28.0 (under cap)
  Net outflows = $110.0 - $28.0               = $82.0

LCR = HQLA / net outflows = $100.0 / $82.0    = 121.95%
```

The bank posts an LCR of about 122%, clearing the 100% floor with roughly $18bn of surplus buffer. Treasury would still watch the trend: a slide toward 100% signals either a thinning buffer or heavier assumed outflows, and much of that HQLA may already be pledged.

NSFR — the one-year structural test

The LCR looks 30 days out; the NSFR looks a full year out and checks the *structure* of funding rather than a short cash sprint. It stops a bank funding a 10-year mortgage book with overnight repo — a mismatch the LCR alone would miss.

> Net Stable Funding Ratio (NSFR) — Available Stable Funding (ASF) ÷ Required Stable Funding (RSF) ≥ 100%. ASF weights liabilities and equity by how reliably they stay for a year; RSF weights assets by how much stable funding they need. The ratio forces long or illiquid assets to be backed by long or stable money.

Item	ASF factor	Item	RSF factor
Capital & >1y debt	100%	Cash & central-bank reserves	0%
Stable retail deposits	95%	Level 1 HQLA (unencumbered)	5%
Less-stable retail	90%	Level 2A HQLA	15%
Operational / non-fin wholesale <1y	50%	Loans to non-fin <1y / Level 2B	50%
Funding from financials <6m	0%	Residential mortgages / loans >1y	65%
—	—	Other loans / illiquid assets >1y	85–100%

WORKED EXAMPLE

A full NSFR calculation. Figures in $bn.

```
AVAILABLE STABLE FUNDING (ASF)
  Equity + >1y debt   $110 × 100%            = $110.0
  Stable retail dep.  $200 × 95%             = $190.0
  Less-stable retail   $50 × 90%             = $45.0
  Op/corporate <1y    $160 × 50%             = $80.0
  Financial <6m        $80 × 0%              = $0.0
  Total ASF                                  = $425.0

REQUIRED STABLE FUNDING (RSF)
  Cash & reserves      $60 × 0%              = $0.0
  Level 1 HQLA        $120 × 5%              = $6.0
  Level 2A HQLA        $20 × 15%             = $3.0
  Mortgages >1y       $300 × 65%             = $195.0
  Other loans >1y     $220 × 85%             = $187.0
  Illiquid other       $30 × 100%            = $30.0
  Total RSF                                  = $421.0

NSFR = ASF / RSF = $425.0 / $421.0           = 100.95%
```

The bank clears the 100% NSFR floor, but only just — at about 101% it has almost no structural headroom. Note the asymmetry treasurers obsess over: a reverse repo against non-HQLA attracts RSF while the matching repo gives little ASF, which quietly makes low-margin matched-book repo expensive to run. To widen the margin the bank would issue more term debt (raising ASF) or shift the asset mix toward shorter, more liquid holdings (lowering RSF).

BEST IN CLASS

Top-tier liquidity teams manage the LCR and NSFR as a matched pair — the 30-day sprint and the one-year marathon — and never optimise one at the other's expense. They keep the HQLA buffer heavily weighted to unencumbered Level 1, because they know that selling or repo-ing the buffer in a stress moves the price and signals distress, so usable liquidity is always less than the headline stock. They scrutinise every run-off and inflow assumption rather than accepting the regulatory minimums as truth, and they model the NSFR asymmetries (matched-book repo, undrawn lines, derivative collateral triggers) that quietly consume stable funding. They issue term debt and gather sticky deposits ahead of need, not when the ratio is already slipping. The decisive habit: they report to ALCO not the regulatory ratio alone but the *usable* liquidity after encumbrance — the number that actually determines whether the bank survives a bad Monday.

07 · COLLATERAL & MARGIN

Collateral & Margin Management

Collateral is the plumbing that turns a promise into security. It converts an unsecured derivative exposure into a secured one, and in doing so it reshapes a bank's funding, its balance sheet, and the price it can quote. This chapter walks from the governing legal documents through the mechanics of margin, the regulatory regime, and the quantitative models desks run every day.

Why collateral exists

Two firms enter a derivative. The contract has a mark-to-market value that moves every day. If the trade gains value for you, your counterparty owes you that gain; if the counterparty defaults before paying, you lose it. This is counterparty credit risk. Collateral removes most of it: the counterparty posts assets as security, and if it fails you keep them. The exposure becomes secured rather than unsecured.

Collateral does not delete risk from the system; it transfers and reduces it. One firm's unsecured credit exposure becomes a secured exposure whose quality now depends on the collateral itself — its liquidity, its price stability, and whether you can actually take and sell it in a default.

> ISDA Master Agreement — the master legal contract between two derivatives counterparties, published by the International Swaps and Derivatives Association. It governs every trade between the pair and defines default events, early termination, and close-out netting — the right to collapse many trades into a single net amount on default.

> Credit Support Annex (CSA) — the collateral rulebook annexed to the ISDA Master. It sets which assets are eligible, how often collateral is exchanged, the thresholds, the haircuts, the minimum transfer amount, and the dispute process.

Variation Margin versus Initial Margin

Collateral comes in two forms that cover two different risks.

> Variation Margin (VM) — collateral that covers the current mark-to-market move. Each day the trades are re-valued and the losing party pays VM to the winning party. VM tracks today's exposure and flows both ways over the life of the relationship.

> Initial Margin (IM) — a buffer posted up front against potential future exposure during the close-out period — the days needed to replace a defaulted counterparty's trades (regulators assume a 10-business-day margin period of risk for uncleared trades, 5 for cleared). IM is held, segregated, and not netted daily.

Put simply: VM pays for the loss that has already happened; IM pre-funds the loss that could happen while you unwind after a default. VM settles daily; IM sits as a locked, segregated buffer.

The CSA parameters

A CSA sets the exact numbers that determine how much collateral actually moves. Learn these terms precisely — they are the levers a treasury negotiates.

Term	Meaning
Threshold	An amount of unsecured exposure allowed before any collateral is due. Below it, no call.
Minimum Transfer Amount (MTA)	The smallest call worth making, to avoid tiny, costly transfers. A typical MTA is $250k–$500k.
Independent Amount (IA)	An extra cushion over and above VM, posted up front — the bilateral precursor to regulatory IM.
Haircut	A discount on posted non-cash collateral. A bond worth $100 at a 2% haircut counts as $98; the haircut absorbs the asset's own price risk.
Eligibility schedule	The list of assets each party may post — e.g. cash in major currencies and G7 government bonds.
Concentration limit	A cap on how much of one issuer or asset type may be posted, preventing over-reliance on one security.

WORKED EXAMPLE

Sizing a VM call after a mark-to-market move. You hold a swap portfolio against a counterparty under a one-way CSA with the following terms:

```
Mark-to-market exposure to you    = $7.30m
Threshold (unsecured allowance)   = $5.00m
Minimum Transfer Amount (MTA)     = $0.50m
Collateral already held           = $0.00m
```

Step 1 — subtract the threshold. Collateral is due only on exposure above it:

```
$7.30m - $5.00m = $2.30m required
```

Step 2 — subtract collateral already held:

```
$2.30m - $0.00m = $2.30m shortfall
```

Step 3 — test against the MTA:

```
$2.30m > $0.50m → the call is valid
```

Result. You call $2.30m of VM. Had the shortfall been below $0.50m you would call nothing. If the counterparty delivers a government bond carrying a 2% haircut rather than cash, it must gross up for the haircut: to deliver $2.30m of collateral *value* it posts $2.30m ÷ 0.98 ≈ $2.347m of bonds. The next day, if your exposure drops to $6.00m, the required collateral falls to $1.00m and — netting the $2.30m held — a *return* of $1.30m is owed back to the counterparty, again subject to the MTA.

Cleared, bilateral and tri-party

A derivative is settled in one of two ways. Either a central counterparty stands in the middle, or the two firms face each other directly.

> Central counterparty (CCP) — a clearing house that steps into the middle of a trade by novation, replacing one contract between A and B with two contracts — A faces the CCP, B faces the CCP. It becomes buyer to every seller and seller to every buyer. Members post IM and VM to the CCP and contribute to a mutualised default fund.

On a member default the CCP runs a default waterfall: (1) the defaulter's own IM, (2) the defaulter's default-fund contribution, (3) a slice of the CCP's own capital ("skin in the game"), then (4) surviving members' default-fund contributions. Bilateral trades under a CSA have no mutualised fund; each firm carries the other's

credit risk net of collateral held. Post-2008, the Dodd-Frank Act (US) and EMIR (EU) impose a clearing mandate pushing standardised products — most vanilla interest-rate swaps, index CDS — into CCPs.

> Tri-party collateral management — a neutral agent (BNY, J.P. Morgan, Euroclear, Clearstream) holds both sides' assets and automates daily movements. It performs auto-allocation: each day it selects eligible assets from the poster's account, applies the correct haircuts and concentration limits, and moves them to the receiver's segregated account, removing manual settlement and reducing disputes.

Rehypothecation and reuse

> Rehypothecation — the reuse of collateral you have received: you take a bond as margin, then post that same bond elsewhere or lend it out. Collateral velocity measures how many times one asset is reused along a collateral chain.

Reuse is efficient — one pool of high-quality assets supports many obligations — but it links firms together. When Lehman Brothers failed in 2008, client assets it had rehypothecated were trapped in the estate; hedge funds that had permitted reuse became unsecured creditors and waited years to recover. The lesson drove a central feature of the modern regime: regulatory IM must be segregated and cannot be rehypothecated.

Uncleared Margin Rules and the AANA threshold

Not every derivative can clear; bespoke trades stay bilateral. For these, regulators imposed the Uncleared Margin Rules (UMR), forcing two-way margin on uncleared OTC derivatives.

> AANA (Aggregate Average Notional Amount) — the measure UMR uses to pull firms into scope. Phase-in ran by size as the AANA threshold fell: Phase 1 (2016) captured dealers above ~$3trn; the final Phase 6 (September 2022) pulled in many smaller buy-side firms at an AANA near €8bn / $8bn.

Under UMR both parties post regulatory IM (reg-IM), two-way, segregated at a third-party custodian and not reusable. VM is exchanged and used; reg-IM is frozen. Most firms compute reg-IM using SIMM.

> ISDA SIMM (Standard Initial Margin Model) — a common, risk-based sensitivity model. IM is built from delta, vega and curvature sensitivities, each multiplied by a prescribed risk weight, then aggregated within and across risk "buckets" using prescribed correlations. A shared model minimises IM disputes because both parties compute the same number.

SIMM aggregation within a risk class follows the quadratic form $IM = \sqrt{(\Sigma K^2 + \Sigma\Sigma\ \rho \cdot K_i \cdot K_j)}$, where each K is a weighted-sensitivity bucket margin and ρ the prescribed inter-bucket correlation. Delta, vega and curvature margins are computed separately per risk class, summed, and then the six risk classes (rates/FX, credit-qualifying, credit-non-qualifying, equity, commodity, FX) are combined with a further correlation matrix.

WORKED EXAMPLE

Stylised SIMM delta IM on a rates portfolio. A desk holds USD interest-rate swap risk expressed as PV01 (dollar sensitivity to a 1bp move) at three tenors. SIMM assigns each tenor a risk weight (shown in bp-equivalent terms) and a within-bucket correlation ρ = 0.98 between adjacent rate tenors (we use a single ρ for simplicity).

Tenor	PV01 ($/bp)	Risk weight (bp)	Weighted sens. WS_k ($)
2y	+40,000	53	2,120,000
5y	−25,000	52	−1,300,000
10y	+15,000	52	780,000

Step 1 — weighted sensitivities $WS_k = PV01_k \times \text{risk weight}_k$ (already in the table).

Step 2 — aggregate the bucket with the SIMM form $K_{rates} = \sqrt{(\Sigma\, WS_k^2 + \Sigma_{i \neq j}\, \rho \cdot WS_i \cdot WS_j)}$:

```
Σ WSₖ²   = 2,120,000² + (-1,300,000)² + 780,000²
         = 4.4944e12 + 1.6900e12 + 0.6084e12
         = 6.7928e12

cross-terms (ρ = 0.98), each ordered pair counted once ×2:
 2y·5y : 0.98 × 2,120,000 × (-1,300,000) = -2.7013e12
 2y·10y: 0.98 × 2,120,000 ×   780,000    = +1.6205e12
 5y·10y: 0.98 × (-1,300,000) × 780,000   = -0.9937e12
 sum of pairs = -2.0745e12 ; ×2 = -4.1490e12

Krates = √(6.7928e12 - 4.1490e12)
       = √(2.6438e12)
       = $1,626,000  (≈ $1.63m)
```

Result. The delta IM for this rates bucket is ≈ $1.63m. Note how the negative 5y position hedges the long 2y/10y risk: the gross weighted sensitivities sum to $1.60m in absolute terms far above the net, yet the correlated aggregation collapses them to $1.63m — barely above the dominant single bucket — because SIMM rewards offsetting directional risk. Vega and curvature margins would be computed the same way and added; cross-currency or multi-asset portfolios then combine risk classes with the top-level correlation matrix.

Collateral optimisation and cheapest-to-deliver

A large firm holds many assets and faces many calls. Which asset should it post where? Posting the wrong one is expensive. The governing rule is cheapest-to-deliver: post the asset least useful elsewhere that is still eligible, and keep your scarce, liquid, high-return assets free. An eligibility-aware optimiser searches every CSA and

CCP rulebook and allocates the optimal asset to each call, ranked down a collateral waterfall — abundant low-yield cash first, then spare government bonds, with scarce HQLA and high-return assets posted last.

WORKED EXAMPLE

Haircut-adjusted posting across two venues. A dealer owes IM of $50m to a CCP (accepts cash at 0% haircut and USTs at 2%) and $30m to a bilateral counterparty (accepts USTs at 2% and investment-grade corporates at 8%). It holds $20m spare cash earning ~0%, $70m of USTs it also wants in its LCR buffer, and $40m of corporates.

```
CCP call $50m: post $20m cash (value $20m) + USTs for $30m value
               USTs needed = $30m ÷ 0.98 = $30.61m face
Bilateral $30m: corporates are least useful for LCR → post corporates
               corporates needed = $30m ÷ 0.92 = $32.61m face
UST face left in buffer = $70m - $30.61m = $39.39m (stays LCR-eligible)
```

Result. By steering cash and corporates to the calls and sparing the most USTs, the dealer meets $80m of margin while keeping $39.39m of HQLA counting toward its LCR — the lowest-opportunity-cost allocation.

BEST IN CLASS

A best-in-class collateral function runs a single enterprise inventory of every eligible asset, netted across all CSAs, CCPs and repo desks in real time; it treats cheapest-to-deliver as a managed P&L with a daily funding-cost benchmark, routes every call through an eligibility- and haircut-aware optimiser, and never posts HQLA where a cheaper eligible asset exists. It reconciles disputes same-day on a shared SIMM calculation, segregates reg-IM with daily custodian attestation, and reports collateral velocity and encumbrance to ALCO as first-class liquidity metrics rather than back-office statistics.

08 · CAPITAL & THE BALANCE SHEET

Capital, Leverage & the Balance Sheet

Every trade consumes a scarce regulatory resource before it earns a cent. This chapter builds the capital stack from the ground up — risk-weighted assets, CET1, the leverage and supplementary leverage ratios, the G-SIB surcharge — and shows how treasury "rents" the balance sheet to desks and judges whether a trade is worth the resource it burns.

Risk-weighted assets

> Risk-weighted assets (RWA) — a bank's assets scaled by their riskiness. A $100m loan to a highly rated sovereign might carry a 0% risk weight (0 RWA); the same amount lent unsecured to a corporate might carry 100% (full $100m RWA). Capital requirements are set as a percentage of RWA, so riskier assets demand more capital.

RWA has three pillars under Basel III: credit RWA (default risk of loans, bonds, counterparty exposure on derivatives and repo), market RWA (trading-book price risk, now under the Fundamental Review of the Trading Book), and operational RWA (loss from failed processes, fraud, litigation). Credit RWA dominates for most banks.

The capital stack and CET1

> Common Equity Tier 1 (CET1) — the highest-quality loss-absorbing capital: common shares plus retained earnings, net of regulatory deductions (goodwill, deferred tax assets). It sits at the bottom of the capital stack and absorbs losses first.

Above CET1 sit Additional Tier 1 (AT1) — perpetual instruments that convert or write down in stress — and Tier 2 subordinated debt. Basel III sets a 4.5% CET1 minimum, plus a 2.5% capital conservation buffer, a countercyclical buffer (0–2.5%), and for the largest banks a G-SIB surcharge — so a large bank's effective CET1 requirement is often 10–13%.

WORKED EXAMPLE

CET1 ratio from capital and RWA. A bank reports CET1 capital of $92bn and total RWA of $760bn.

```
CET1 ratio = CET1 capital ÷ RWA
           = 92 ÷ 760
           = 0.1211 = 12.11%
```

Result. A 12.11% CET1 ratio sits comfortably above a ~11% requirement. Now suppose the bank adds $40bn of corporate loans at a 100% risk weight, funded without new equity:

```
new RWA  = 760 + 40 = 800bn
new CET1 = 92 ÷ 800 = 11.50%
```

The same capital now supports more risk; the ratio drops 61bp. To hold 12.11%, the bank would need 0.1211 × 800 = $96.9bn of CET1 — $4.9bn more capital for $40bn of new RWA, i.e. a marginal capital intensity of ~12.1 cents per dollar of RWA. That marginal cost is exactly what treasury charges the lending desk.

The leverage ratio and SLR

RWA can be gamed: pile into low-risk-weight assets and the RWA denominator barely moves even as the balance sheet balloons. The leverage ratio is the non-risk-based backstop.

> Leverage ratio — Tier 1 capital divided by total leverage exposure: on-balance-sheet assets plus off-balance-sheet add-ons (derivatives potential future exposure, securities-financing exposure, commitments), with no risk weighting. Basel sets a 3% minimum.

> Supplementary Leverage Ratio (SLR) — the US implementation, with a 3% minimum. The enhanced SLR (eSLR) adds a 2% buffer for US G-SIB holding companies (5% total) and 6% at their insured bank subsidiaries.

WORKED EXAMPLE

SLR calculation including off-balance-sheet add-ons. A US G-SIB holding company reports:

```
Tier 1 capital                       = $170bn
On-balance-sheet assets              = $2,900bn
Derivatives PFE add-on               = $120bn
Securities-financing (repo) exposure = $180bn
Off-balance commitments (CCF-adj)    = $100bn
-------------------------------------------------
Total leverage exposure = 2,900 + 120 + 180 + 100 = $3,300bn

SLR = Tier 1 ÷ total leverage exposure
    = 170 ÷ 3,300
    = 5.15%
```

Result. At 5.15% the firm clears the 5% eSLR with 15bp — ~$5bn of spare Tier 1 — to spare. A new $100bn matched-book reverse-repo adds ~$100bn of leverage exposure but almost no RWA; it alone drops the SLR to 170 ÷ 3,400 = 5.00%, exhausting the buffer. This is why low-risk, low-margin repo is leverage-expensive even when it is RWA-cheap, and why dealers pull back from repo at quarter-end reporting dates.

The G-SIB surcharge and Method 2

> G-SIB surcharge — extra CET1 required of Globally Systemically Important Banks, scaled to systemic footprint. The Basel/FSB framework scores size, interconnectedness, cross-jurisdictional activity, complexity and substitutability. The US "Method 2" replaces substitutability with a reliance on short-term wholesale funding term, typically producing a higher, stickier surcharge (buckets of 1.0%–3.5% CET1).

Because Method 2 penalises short-term wholesale funding and balance-sheet size, it directly shapes treasury behaviour: it rewards term funding and stable deposits, and it makes end-of-year balance-sheet "window dressing" a genuine capital lever, since the surcharge is calibrated off year-end data.

Renting the balance sheet

A trading desk does not own capital or leverage capacity — it borrows them from the firm. Treasury (via FTP and the XVA desk) charges each desk for the scarce resource its trades consume, so that the firm deploys capital where it earns most. Two return metrics govern this allocation:

- Return on RWA (RoRWA) — trade P&L divided by the RWA it generates; the yardstick for risk-weighted businesses.

- Return on balance sheet / return on leverage exposure — P&L divided by the leverage exposure consumed; the binding yardstick for repo and low-risk books where the leverage ratio bites before RWA does.

RWA and SLR can be optimised by the same levers that run collateral: netting (close-out and payment netting shrink gross exposure), trade compression (tearing up offsetting trades reduces gross notional and leverage exposure), and collateral (posting/receiving eligible collateral reduces counterparty credit RWA via lower exposure-at-default).

WORKED EXAMPLE

Balance-sheet "rental" charge and return-on-RWA of a repo book. A financing desk runs a $5bn matched reverse-repo/repo book earning a 6bp net spread. Counterparty credit RWA on the book is computed at $250m; leverage exposure is $5bn. Treasury's hurdles: 12% cost of equity on an 11% CET1 charge against RWA, plus a leverage-capacity rent of 20bp on leverage exposure.

```
Gross P&L  = $5,000m × 0.0006 = $3.0m / yr

RWA capital charge = RWA × CET1% × CoE
                   = 250 × 0.11 × 0.12 = $3.3m...
   (per $m: 250m RWA ×11% =$27.5m equity × 12% = $3.30m)
Leverage rent      = 5,000 × 0.0020 = $10.0m  (illustrative gross bp)
   → realistic leverage rent at 2bp = 5,000 × 0.0002 = $1.0m

Economic P&L = 3.0 - (capital 3.3 scaled) ...
```

Result. Take the realistic charges: capital cost on $250m RWA at 11% CET1 and a 12% cost of equity = 250 × 0.11 × 0.12 = $3.30m, and a leverage rent of 2bp on $5bn = $1.0m. Against $3.0m of gross spread the book loses money: 3.0 − 3.30 − 1.0 = −$1.30m economic P&L. Its RoRWA is 3.0 ÷ 250 = 1.2%, far below a ~13% hurdle. The desk must either widen the spread, compress the book to cut RWA and leverage exposure, or hand the capacity to a higher-returning business. This is treasury renting the balance sheet in action — the trade looked profitable on spread alone and was destroying value once the resource was priced.

BEST IN CLASS

A best-in-class capital function gives every desk a live, trade-level view of the RWA, leverage exposure, G-SIB-score and capital charge each ticket consumes, and prices all of them into the hurdle at deal time — not in a quarterly post-mortem. It runs standing compression cycles with TriOptima/CCP services, manages SLR headroom against reporting-date spikes rather than being surprised by them, and treats the G-SIB Method-2 score as an optimisable quantity (terming out wholesale funding, steering size off measurement dates). RoRWA and return-on-leverage are reported side by side so the genuinely binding constraint — risk-based or leverage-based — governs every allocation.

09 · XVA

XVA — Pricing the Balance Sheet into a Trade

The clean, textbook price of a derivative is only a starting point. Real desks adjust it for default risk, funding, margin, collateral and capital — the XVA family — turning a risk-free value into the all-in price a client is actually charged. This chapter defines each adjustment, its sign, and shows how a quoted price is built up.

The valuation-adjustment family

XVA is a group name; the "X" is a placeholder. Each letter is one adjustment that corrects the clean price for one real cost or benefit.

Adj.	Stands for	What it prices	Sign
CVA	Credit Valuation Adj.	Expected loss if the counterparty defaults	− cost
DVA	Debit Valuation Adj.	Expected gain if *you* default and do not pay	+ benefit
FVA	Funding Valuation Adj.	Cost/benefit of funding an uncollateralised position	±
MVA	Margin Valuation Adj.	Cost of funding posted initial margin	− cost
KVA	Capital Valuation Adj.	Cost of holding regulatory capital over the trade's life	− cost
ColVA	Collateral Valuation Adj.	Rate and currency effects of the collateral itself	±

> XVA — the full set of valuation adjustments applied to a derivative. The all-in price is the clean (risk-free, perfectly collateralised) price minus costs plus benefits: Price = Clean − CVA + DVA − FVA − MVA − KVA ± ColVA.

Exposure: the raw material of CVA

> Expected Positive Exposure (EPE) — the average of the trade's future value *when that value is positive to you* (i.e. the counterparty owes you), measured at each future date across thousands of simulated market paths. Only positive exposure matters, because you only lose on default when the counterparty owes you.

> Potential Future Exposure (PFE) — a high percentile (e.g. 95th or 99th) of future exposure — the plausible worst case, used for limits and IM sizing rather than for CVA pricing.

CVA integrates expected exposure against the counterparty's probability of default and loss given default: CVA ≈ LGD × Σ EPE_t × marginal PD_t × DF_t, summed over time buckets, where marginal PD is the chance of default in each interval (bootstrapped from the counterparty's CDS spread) and DF is the discount factor.

WORKED EXAMPLE

CVA on an uncollateralised 3-year swap. A simplified exposure profile and credit inputs:

Year	EPE_t ($m)	Marginal PD_t	DF_t	Contribution ($m)
1	2.0	1.60%	0.962	0.0308
2	3.0	1.57%	0.925	0.0436
3	2.5	1.55%	0.889	0.0344
Σ EPE·PD·DF				0.1088

```
LGD = 60% (recovery 40%)
CVA = LGD × Σ (EPEₜ × marginal PDₜ × DFₜ)
    = 0.60 × 0.1088
    = $0.0653m = $65,300
```

Where the PDs come from: a flat CDS spread of ~100bp with 60% LGD implies a hazard rate λ ≈ spread/LGD ≈ 0.0100/0.60 ≈ 1.67% p.a.; the marginal (interval) default probabilities above are the survival-weighted differences, declining slightly each year as cumulative survival falls.

Result. The CVA charge is ≈ $65,300. On a swap whose clean value to the dealer is, say, $120,000, CVA alone consumes more than half the apparent profit — and it would double if the counterparty's CDS doubled, which is why CVA is hedged with CDS and index protection.

FVA and the funding debate

A fully collateralised trade funds itself: cash collateral flows in exactly as value moves, so no FVA arises. An uncollateralised trade is different — the dealer must borrow cash to carry it at its own funding spread above the risk-free rate, a real cost over the trade's life.

> FVA (Funding Valuation Adjustment) — the present value of funding an uncollateralised exposure. It nets a Funding Cost Adjustment (FCA) — the cost of funding shortfalls when the trade is in-the-money to the dealer — against a Funding Benefit Adjustment (FBA) — the saving when the dealer holds reusable cash.

FVA was contentious. From 2013–2014, banks took large one-off accounting hits as they adopted it — JPMorgan's ~$1.5bn FVA write-down in Q4 2013 was the landmark, with similar charges at other dealers. The

controversy is the FVA–DVA overlap: both relate to the dealer's own credit, so booking a full DVA benefit *and* a full FBA risks counting the same funding gain twice. Desks must define and net the terms carefully.

WORKED EXAMPLE

FVA from a funding spread. An uncollateralised trade is expected to run an average net funding requirement (expected positive exposure the dealer must fund) of $3.0m over a 4-year life. The dealer's funding spread over the risk-free/OIS curve is 90bp.

```
Annual funding cost = expected funded amount × funding spread
                    = 3.0m × 0.0090 = $27,000 / yr

Over 4 years, discounted at ~4% (annuity factor ≈ 3.63):
FVA ≈ 27,000 × 3.63 = $98,000  (≈ $0.098m)
```

Result. The FVA charge is ≈ $98,000. If this dealer's funding spread were only 40bp (a stronger-rated bank), the same trade's FVA would be $3.0m × 0.0040 × 3.63 ≈ $43,600 — less than half. Funding quality is a direct competitive weapon: the better-funded bank can quote a keener price for identical risk.

Building the all-in price

WORKED EXAMPLE

Clean price to client quote. Combining the two trades above on a single uncollateralised deal with a clean (risk-free) value to the dealer of $120,000, add the MVA and KVA the XVA desk computes:

```
Clean price (dealer's favour)       +120,000
 - CVA   (counterparty default)      - 65,300
 + DVA   (own default benefit)       + 18,000
 - FVA   (funding, net of DVA ovlp)  - 85,000
 - MVA   (IM funding over life)      - 22,000
 - KVA   (capital over life)         - 31,000
-----------------------------------------------
 All-in value to dealer             - 65,300 → -$65.3k
```

Result. The clean price showed a $120,000 gain, but once every scarce resource is priced the trade is *underwater by ~$65,000* at mid. To clear its hurdles the desk must charge the client roughly $185,000 more than the clean mid — the XVA markup — or restructure the trade (add a CSA, clear it, or net it against existing exposure) to shrink CVA, FVA and MVA. This is why the same economic swap is quoted at wildly different spreads to a well-collateralised dealer versus an uncollateralised corporate.

BEST IN CLASS

A best-in-class XVA capability runs a single central desk that prices CVA, DVA, FVA, MVA, KVA and ColVA on every new ticket at deal time, charges the originating desk transparently, and actively hedges the residual CVA and funding risk across the whole book rather than per trade. It uses one consistent exposure engine (same Monte-Carlo paths for EPE, PFE and IM), nets the FVA–DVA overlap explicitly with a documented methodology, and feeds live XVA sensitivities to the salesforce so a trader can see instantly whether adding a CSA, clearing, or netting against existing exposure turns a value-destroying quote into a winnable one.

10 · INTRADAY, PAYMENTS & CRISES

Intraday Liquidity, Payments & Crises — War Stories

Liquidity does not fail at end of day; it fails at 11am when a payment is due and the cash is not there. This chapter covers intraday liquidity management and the payment plumbing that carries it, then runs a treasury-lens post-mortem of five crises — what broke, the figures, and the lesson a treasurer carries forward.

Intraday liquidity management

A bank's liquidity position swings violently within a single day as payments arrive and leave. Managing the end-of-day balance is not enough; the bank must have cash *at the moment* each obligation settles.

> Intraday liquidity — the funds a bank needs during the business day to meet payment and settlement obligations as they fall due, before offsetting inflows arrive. A bank can be solvent and end the day flat yet still fail intraday if it cannot fund a mid-morning peak.

> BCBS 248 — the Basel monitoring tools for intraday liquidity (2013). Key metrics: daily maximum intraday liquidity usage (the largest net negative position during the day), available intraday liquidity (central-bank reserves, committed lines, unencumbered HQLA), total payments, and time-specific obligations that must settle by a set hour.

Treasury watches throughput (the share of a day's payments completed by set times) and peak usage relative to available resources. A bank that releases payments too late starves the system; one that funds its peak too conservatively ties up expensive HQLA. The balance is a daily optimisation.

Payment systems and settlement

> RTGS (Real-Time Gross Settlement) — a system (Fedwire in the US, TARGET2/T2 in the euro area, CHAPS in the UK) that settles each large-value payment individually and with finality in central-bank money, in real time. "Gross" means no netting — every payment moves in full, so intraday liquidity demand is high.

> Herstatt risk — settlement risk in FX, named after Bankhaus Herstatt (failed 1974): you pay away one currency, and before you receive the other leg the counterparty fails in a different time zone, so you lose the full principal. It is cross-currency principal risk arising from non-simultaneous settlement.

> CLS (Continuous Linked Settlement) — the multi-currency settlement system that solves Herstatt risk via payment-versus-payment (PvP): both legs of an FX trade settle simultaneously or not at all, so principal is never at risk. CLS settles the large majority of the global FX market across major currencies.

Crisis post-mortems

2008 — the wholesale funding run

Mechanism. Banks and dealers had funded long-dated, illiquid assets with very short-term wholesale money — overnight repo, ABCP, prime money-market-fund paper. When asset quality came into doubt, lenders refused to roll. Bear Stearns' and Lehman's repo books evaporated in days; repo haircuts on private-label collateral jumped from low single digits to 25–45%. Figures. The ABCP market shrank from ~$1.2trn (2007) to a fraction of that; Lehman failed on 15 September 2008 with ~$600bn of assets. Lesson: funding liquidity is a confidence good — it disappears instantly and all at once. This crisis birthed the LCR and NSFR, which force a stock of HQLA and stable term funding against short-dated reliance.

September 2019 — the repo spike

Mechanism. A corporate-tax payment date and a large Treasury settlement drained reserves on the same day, just as the Fed's balance-sheet runoff had pushed bank reserves to their scarce floor. Dealers constrained by SLR could not expand their repo books to intermediate. Figures. Overnight GC repo and SOFR spiked from ~2.2% to a ~10% intraday high on 17 September 2019; the Fed restarted repo operations and reserve purchases. Lesson: "excess" reserves are not fungible with usable intraday liquidity when balance-sheet constraints bind; the level of reserves at which the system seizes is higher than anyone assumed.

March 2020 — the dash for cash

Mechanism. A pandemic scramble for dollars forced levered holders to sell even the safest asset — US Treasuries — to raise cash. The Treasury-basis trade (hedge funds long cash Treasuries versus short futures on heavy repo leverage) blew up as rising margin calls forced forced selling, and dealers hit SLR limits could not warehouse the bonds. Figures. Treasury market liquidity collapsed to crisis levels; the Fed bought ~$1trn of Treasuries in weeks and launched facilities (PDCF, MMLF, CPFF). The Fed temporarily excluded reserves and Treasuries from the SLR denominator to free dealer balance sheets. Lesson: even the risk-free asset can become illiquid when everyone needs cash at once; dealer balance-sheet capacity is the binding constraint on market liquidity.

2022 — the UK LDI margin spiral

Mechanism. UK pension funds ran liability-driven investment (LDI) strategies using leveraged gilt positions and interest-rate swaps to hedge long-dated liabilities. When the September 2022 "mini-budget" sent gilt yields soaring, the hedges generated huge VM and repo margin calls. To meet them, funds sold gilts, pushing yields higher still — a self-reinforcing doom loop. Figures. 30-year gilt yields rose ~120bp in days; the Bank of England launched a temporary £65bn gilt-buying backstop. Lesson: a hedge is only as good as your ability to fund its margin; leveraged collateral strategies must hold a liquidity buffer sized for a tail move, not a normal one.

2023 — Silicon Valley Bank

Mechanism. SVB took in a flood of tech deposits and invested them in long-dated fixed-rate Treasuries and MBS, booking much of it as held-to-maturity (HTM) to avoid marking losses through equity. As rates rose sharply through 2022, the portfolio's economic value fell far below book — a classic duration mismatch. Its deposits were overwhelmingly uninsured (above the FDIC $250k cap) and concentrated in one fast-moving community. Figures. SVB held ~$91bn of HTM securities carrying ~$15bn of unrealised losses against ~$16bn of tangible equity; depositors tried to pull ~$42bn in a single day (10 March 2023). The bank failed overnight. Lesson: HTM accounting hides interest-rate risk but not its economics; uninsured, concentrated deposits are the least stable funding there is, and in the smartphone era a run is measured in hours, not days. Duration risk and funding risk are the same risk viewed from two sides of the balance sheet.

WORKED EXAMPLE

SVB's duration mismatch, quantified. Suppose a bank holds $90bn of fixed-rate bonds with a modified duration of 6 years, funded by uninsured deposits. Rates rise 250bp.

```
Price change ≈ -duration × Δyield × value
             ≈ -6 × 0.0250 × 90bn
             ≈ -$13.5bn mark-to-market loss

Against tangible equity of ~$16bn:
 remaining economic equity ≈ 16 - 13.5 = $2.5bn
 → ~84% of equity erased by the rate move alone
```

Result. A 250bp move wipes out ~$13.5bn — nearly all the bank's capital — even though the bonds are "safe" and credit-risk-free. Held to maturity the loss never hits reported equity; but the moment a deposit run forces a sale, the HTM fiction collapses into a realised loss and the bank is insolvent. This is the mechanism BCBS 248, the LCR and interest-rate-risk-in-the-banking-book (IRRBB) rules exist to surface before it is fatal.

BEST IN CLASS

A best-in-class treasury manages intraday liquidity in real time against the BCBS 248 tools, schedules time-critical payments early to maximise throughput, and holds a buffer sized to the largest observed intraday peak rather than the end-of-day balance. It settles FX through CLS to eliminate Herstatt risk, stress-tests its collateral and margin needs against a 2019/2020-scale tail (not a normal day), runs its HTM/AFS book inside explicit IRRBB and duration limits with the economic loss reported to ALCO monthly, and treats deposit concentration and insured-versus-uninsured mix as a monitored, limited risk — because every crisis in this chapter was, at root, stable-looking funding proving unstable at the worst possible moment.

11 · LIQUIDITY CRUNCHES

A Century of Liquidity Crunches — Patterns & Calculations

Every liquidity crunch in the last hundred years rhymes. The names, the instruments and the acronyms change — call loans, commercial paper, repo, money-market funds, LDI, stablecoins — but the underlying machine is identical. This chapter lays out the repeating seven-stage anatomy, catalogues the major crunches from 1907 to 2023, and then does the thing practitioners actually need: the arithmetic of how a buffer drains, how a haircut forces a fire sale, and how a doom loop feeds itself.

The universal anatomy of a liquidity crunch

Strip away the era and the instrument, and a liquidity crunch is a seven-stage sequence. It is worth memorising because recognising which stage you are in tells you which actions are still available. Once you are past stage four, you are almost always a price-taker.

1. A mismatch builds in calm times. An institution funds long, illiquid or hard-to-value assets with short, runnable money. In the quiet, this is simply profitable: the funder earns the term-and-liquidity premium. The mismatch is invisible precisely because nothing has gone wrong yet.
2. A trigger cracks confidence. A loss, a downgrade, a failed auction, a counterparty default, a rumour. The trigger rarely needs to be large relative to the balance sheet; it only needs to make the short-term funders ask whether they will be repaid.
3. Short-term funders run. Depositors withdraw, repo lenders refuse to roll or demand more collateral, commercial-paper buyers stay home, fund investors redeem. Whoever can leave cheapest and fastest leaves first — and that is almost always uninsured wholesale money.
4. Forced asset sales at fire-sale prices. To meet the outflow the firm sells assets. Because everyone holding the same assets is doing the same thing, the only clearing price is well below fundamental value.
5. Mark-to-market losses and margin/haircut spirals amplify. The fire-sale price becomes *the* price. It marks down everyone else's book, triggers more margin calls, and lenders raise haircuts — so the same position now needs more equity exactly when equity is scarcest.
6. Contagion spreads. Through common asset exposures (everyone marked to the same fire sale), through direct counterparty chains, and through pure information (if that firm, then why not this one?).
7. A backstop arrives — or the firm fails. A lender of last resort lends freely against good collateral, a dealer of last resort buys the assets outright, or a government guarantees the funding. Absent a backstop large and fast enough, the firm fails and the cycle repeats elsewhere.

> Fire sale — the forced sale of assets into a market where most natural buyers are themselves distressed, so the clearing price falls far below fundamental value; the discount is a liquidity premium, not a solvency judgement.

> Margin spiral — falling prices trigger variation-margin calls; meeting the calls requires selling, which pushes prices down further, which triggers more calls. A self-reinforcing loop on the *price* axis.

> Haircut spiral — as volatility rises, lenders raise the haircut on collateral; higher haircuts force deleveraging (you can borrow less against the same bond), which forces sales, which raises volatility, which raises haircuts. A self-reinforcing loop on the *leverage* axis. Brunnermeier and Pedersen call the pair “loss spirals” and “margin spirals”.

> Flight to quality / dash for cash — investors simultaneously rotate out of risky and even high-quality-but-illiquid assets into the safest, most liquid instrument — cash and T-bills. In March 2020 even on-the-run Treasuries were sold to raise dollars.

> Contagion — the transmission of distress from one institution or market to others via common exposures, counterparty credit chains, and confidence effects, even when the second institution was fundamentally sound.

> Doom loop — a two-sided feedback where distress in A worsens B, and distress in B worsens A. The canonical pair is a sovereign and its banks; the 2022 LDI episode was a pension-fund/gilt doom loop.

> Lender of last resort (LOLR) — Bagehot's rule: in a panic the central bank should lend freely, against good collateral, at a penalty rate, to solvent-but-illiquid firms. The modern variants add a *dealer of last resort* (outright asset purchases) and *guarantor of last resort* (funding/deposit guarantees).

The repeating patterns — what every crunch has in common

If you only remember one sentence from this chapter, make it this: liquidity is not solvency, but a long-enough liquidity crisis manufactures insolvency. A firm can be solvent on a mark-to-maturity basis and still die because it cannot meet cash outflows this week. The forced sales then crystallise losses that genuinely destroy the capital that was, an hour earlier, adequate.

Across a century the recurring signatures are strikingly consistent:

- Short, runnable funding against long, illiquid assets. The original sin. The shorter and more concentrated the funding, the faster the run.
- Hidden or pro-cyclical leverage. Leverage that looks modest in calm markets explodes as haircuts rise and asset values fall.
- Crowded, correlated trades. Diversification assumptions that hold in normal times break in stress — correlations go to 1 exactly when you need them not to.

- Collateral pro-cyclicality. Haircuts and initial margin rise in stress, draining funding precisely when it is scarcest.
- Wholesale/uninsured money flees first. Insured retail deposits are sticky; uninsured corporate deposits, repo and CP are not. The funding mix is destiny.
- The “30-day buffer” illusion. Regulatory buffers (LCR) assume a stylised 30-day stress. Real runs can exhaust a buffer in a single afternoon (SVB) because digital, concentrated depositors move together and faster than the rulebook imagines.
- Concentration and interconnectedness. Concentrated depositor bases, concentrated counterparties, and dense interbank links turn an idiosyncratic shock into a systemic one.

Recurring signature	Mechanism	Early tell for a treasurer
Maturity mismatch	Long assets, short funding	Rising share of overnight / <1-week liabilities
Pro-cyclical leverage	Haircuts fall in boom, rise in bust	Unusually thin haircuts on your financing
Correlated/crowded trade	Everyone holds the same position	Your “edge” is everyone's edge
Wholesale funding reliance	Non-sticky money leaves first	Insured-deposit share falling
Concentration	Few depositors / counterparties	Top-20 depositors > large % of total
Collateral quality drift	Lower-grade collateral pledged	Encumbrance ratio rising

A chronological field catalogue

The table below is the century at a glance. Read each row as one instance of the same machine: a mismatch, a trigger, a run, a backstop.

Year	Event	The mismatch	The trigger	Run / loss metric	Backstop
1907	Knickerbocker panic	Trust co. demand deposits vs stock-speculation loans	Failed corner in United Copper	NYC trusts lost ~48% of deposits in weeks	J.P. Morgan-led private liquidity pool
1929–33	Depression bank runs	Demand deposits vs illiquid loans	Crash, waves of bank failures	~9,000 banks failed; money supply −1/3	Bank holiday; FDIC (1933)
1970	Penn Central CP crisis	Short CP rolling long rail assets	Penn Central bankruptcy; CP default	CP market froze; rollovers failed	Fed discount window opened to banks
1984	Continental Illinois	~75% wholesale/ brokered funding	Bad energy loans; telex run	~$8bn fled; largest US bank failure then	FDIC open-bank assistance; “too big to fail”
1987	Black Monday	Portfolio-insurance selling; margin	−22.6% Dow in one day	Clearing/margin near-gridlock	Fed liquidity assurance
1997	Asian crisis	$ short-term debt vs local assets	Thai baht de-peg	Reserves exhausted; capital flight	IMF programmes
1998	LTCM / Russia	~25–100× leverage on convergence trades	Russian default	~$4.6bn capital wiped in weeks	Fed-organised dealer recapitalisation
2007	Northern Rock	Securitisation/ wholesale funded mortgages	Securitisation market froze	First UK bank run since 1866	BoE emergency loan; deposit guarantee
Mar 2008	Bear Stearns	Overnight repo vs illiquid MBS	Repo lenders pulled; hedge funds fled	Cash fell ~$18bn→$2bn in days	Fed-backed JPMorgan takeover
Sep 2008	Lehman / Reserve Primary	Repo & CP vs illiquid assets; MMF held Lehman CP	Lehman bankruptcy	Fund “broke the buck”; ~$300bn MMF run	Fed AMLF; Treasury MMF guarantee
2011–12	Eurozone crisis	Banks ↔ sovereign doom loop	Greek/peripheral debt fears	Peripheral funding froze	ECB LTRO; “whatever it takes” / OMT
2013	Taper tantrum	Levered EM & duration carry	Fed taper signal	10y UST +~100bp; EM outflows	Fed communication walk-back
2015	SNB CHF un-peg	Short-CHF carry; broker leverage	SNB drops the 1.20 floor	EUR/CHF −~30% in minutes; FX brokers blew up	None for brokers; SNB policy shift only
Sep 2019	US repo spike	Reserves scarce vs settlement demand	Tax date + UST settlement drain	SOFR spiked to ~10% intraday	Fed repo ops; later SRF
Mar 2020	COVID dash-for-cash	Everyone short dollars/ liquidity	Pandemic shock	Even on-the-run USTs sold off	Fed unlimited QE, swap lines, facilities
2022	UK LDI gilt crisis	Levered gilt/LDI vs margin calls	“Mini-budget” gilt sell-off	30y gilt yield +~120bp in days	BoE emergency gilt purchases
2023	SVB / Signature / First Republic; Credit Suisse	Uninsured deposits vs HTM bond losses	Capital raise failed; social-media run	SVB: ~$42bn requested in one day	FDIC; Fed BTFP; UBS–CS deal
2022–23	Crypto: Terra / FTX / USDC	Reflexive pegs; commingled client funds	UST de-peg; FTX insolvency; SVB spillover	USDC briefly ~$0.88; Terra →~0	None for Terra/FTX; USDC restored on bank backstop

Chapter 10 already deep-dived 2008 (Lehman and the broader GFC), the September 2019 repo spike, the March 2020 dash-for-cash, the 2022 UK LDI episode, and the 2023 SVB failure. So the mini-cases and calculations below focus on the *shared mechanics* — the money-fund break, the haircut spiral, the survival-horizon arithmetic, the LDI doom loop, and the deleveraging multiplier — because the mechanics transfer across every row of the table.

The mechanics, as calculations

A pattern you cannot compute is a pattern you do not really understand. Here are the five calculations that recur in every crunch. Figures are illustrative teaching numbers grounded in the real events.

1 · A money-market fund “breaks the buck”

Prime money-market funds promise a stable $1.00 net asset value (NAV) per share. They hold short, high-grade paper and amortise it at cost, so the NAV only moves if an asset is impaired. The Reserve Primary Fund held Lehman commercial paper; when Lehman failed on 15 September 2008, that paper was marked toward zero.

WORKED EXAMPLE

```
Fund size (assets)                 = $62.6 bn
Shares outstanding (≈ $1 each)     = 62.6 bn shares
Lehman CP held                     = $785 m
Assume Lehman CP marked to $0 (write-off of full $785m)

Shadow NAV per share = (Assets - Writeoff) / Shares
                     = ($62.600 bn - $0.785 bn) / 62.6 bn
                     = $61.815 bn / 62.6 bn
                     = $0.98746 per share  ≈ $0.987

Loss per share       = $1.000 - $0.987 = $0.013  (about 1.3 cents)
```

A loss of just over one cent per share drops NAV below $1.00 — the fund “breaks the buck” (Reserve Primary ultimately struck $0.97). The number is tiny; the consequence was not. A sub-$1 NAV means early redeemers get $1.00 and late redeemers get less, so every rational holder redeems first. Investors pulled roughly $300bn from prime money funds in days, freezing the commercial-paper market that funds corporate payrolls. The backstop: the Fed's AMLF (lending to banks to buy ABCP from funds) and a temporary Treasury guarantee of money-fund NAVs. Lesson: a stable-value promise plus a first-mover advantage is a run waiting for a trigger — the same structure re-appeared in the 2023 USDC stablecoin de-peg, which was itself an SVB spillover.

2 · The haircut / margin spiral

A haircut is the lender's equity requirement on a financed position. The maximum leverage you can run is simply the inverse of the haircut.

> Haircut — the percentage of a position's value the borrower must fund with its own equity; the lender advances the rest against the collateral. Maximum leverage = 1 / haircut.

WORKED EXAMPLE

```
Position: $100 of a bond.

Calm-market haircut = 2%
  Equity required   = 2% × $100 = $2
  Borrowed (repo)   = $98
  Max leverage      = 1 / 0.02 = 50×

Stress: haircut jumps 2% → 10%
  Equity now required on the SAME $100 = 10% × $100 = $10
  You still only have $2 of equity.

To restore the position to your available $2 of equity at a 10%
haircut:
  Max position = Equity / haircut = $2 / 0.10 = $20

  So you must shrink the book from $100 → $20
  Forced sale      = $100 - $20 = $80 of bonds dumped
```

Nothing happened to the bond's fundamentals. A haircut moving from 2% to 10% — ordinary in a stress — forces an 80% liquidation of the position. Every levered holder does this at once; the selling is the fire sale; the fire sale raises measured volatility; higher volatility justifies even higher haircuts. That is the haircut spiral, and it is why LTCM (1998), the 2008 repo market, and the 2020 basis-trade unwind all deleveraged violently from what looked like modest triggers.

3 · The wholesale-funding run and survival horizon

The single most useful number a treasurer can carry in their head is the survival horizon: how many days the liquidity buffer lasts against the outflow.

> Survival horizon — liquidity buffer divided by expected net cash outflow per period. A bank with buffer *B* and daily outflow *O* survives *B / O* days before it must sell illiquid assets or seek a backstop.

WORKED EXAMPLE

```
CASE A — SVB-style single-day run (digital, concentrated,
uninsured)
  Total deposits               ≈ $173 bn
  Withdrawn / requested in ONE day ≈ $42 bn
  One-day outflow rate = $42 bn / $173 bn = 24.3%

  A regulatory 30-day LCR stress assumes ~ a few % per day.
  Here ~24% left in a single day → the 30-day buffer logic is
irrelevant.
  Survival horizon at that velocity ≈ 173 / 42 ≈ 4 days of total
deposits,
  but liquid buffer was far smaller → effectively < 1 day.
Failure.

CASE B — Continental Illinois / Northern Rock style wholesale
flight
  Funding mix: 75% wholesale (non-insured, must be rolled daily/
weekly)
  Balance sheet funded assets      = $40 bn (illustrative)
  Wholesale funding                = 0.75 × $40 bn = $30 bn
  Assume wholesale CANNOT be rolled at all once confidence
breaks.

  Refinancing gap = wholesale that won't roll - liquid assets
available
                  = $30 bn - (say) $6 bn of truly liquid assets
                  = $24 bn gap that MUST be met by fire sales or
a backstop
```

The two cases bracket the modern problem. Continental Illinois (1984) and Northern Rock (2007) died because ~three-quarters of their funding was wholesale and simply would not roll — a slow-motion run over days. SVB (2023) shows the new regime: concentrated, uninsured, app-enabled depositors coordinating in hours, so the outflow rate that matters is per-*day*, not per-month. In both regimes the arithmetic is the same — buffer over outflow — and in both the gap had to be closed by a lender/guarantor of last resort.

4 · The LDI collateral-call doom loop (2022)

UK pension funds used liability-driven investment (LDI): leveraged gilt and swap positions to hedge long-dated liabilities. Leverage means a yield move produces a price loss that triggers a collateral call — and meeting the call by selling gilts pushes yields up further.

WORKED EXAMPLE

```
Position: £100 of long gilts, 3× leverage
  Capital (equity) posted = £100 / 3 ≈ £33
  Borrowed / notional exposure financed = £67
  Modified duration of the gilts ≈ 12 years

Price change ≈ - duration × Δyield

STEP 1 — yields rise +100bp (+1.00%)
  Price fall = -12 × 1.00% = -12%  → loss = 12% × £100 = £12
  Capital after loss = £33 - £12 = £21
  Loss as % of capital = £12 / £33 = 36%  → large collateral
call

STEP 2 — yields rise a further +150bp (cumulative +250bp)
  Additional price fall = -12 × 1.50% = -18% → further loss
= £18
  Cumulative loss = £12 + £18 = £30 vs original £33 capital
  Remaining capital = £33 - £30 = £3  (≈ 91% of capital
gone)

The fund must post variation margin NOW or be closed out.
  To raise cash it SELLS gilts → yields rise MORE → bigger
losses on
  the gilts it still holds → bigger calls → more forced
selling …
```

That is the doom loop: a +250bp move against a 12-year-duration, 3×-levered book erases essentially all the capital, and the act of meeting the margin call makes the next margin call worse. Left alone it is explosive, so the Bank of England stepped in as dealer of last resort, buying long gilts to cap yields and break the self-feeding sale — a liquidity intervention, explicitly not monetary easing. Note how three of our universal callouts fire at once here: margin spiral, fire sale, and doom loop.

5 · The constant-leverage deleveraging multiplier

A levered institution that targets a constant leverage ratio must sell a *multiple* of any loss just to stay at target. This is why small losses produce outsized selling and why leverage is pro-cyclical.

WORKED EXAMPLE

```
Start: Assets = $100, Equity = $10, Debt = $90  →
leverage = 100/10 = 10×

Shock: assets fall 5%  → Assets = $95, Equity = $10 -
$5 = $5, Debt = $90
  New leverage = 95 / 5 = 19×  (leverage has nearly
DOUBLED)

To restore 10× leverage on the new $5 of equity:
  Target assets = leverage × equity = 10 × $5 = $50
  Must sell = current assets - target = $95 - $50 =
$45

Compare the selling to the dollar loss:
  Dollar loss = $5
  Forced sale = $45
  Multiplier  = $45 / $5 = 9×
```

A 5% asset loss forces the holder to sell nine times the dollar amount of the loss to hold leverage constant. Adrian and Shin documented exactly this behaviour in dealer balance sheets: leverage rises when asset prices rise and collapses when they fall, so intermediaries are forced buyers in booms and forced sellers in busts. The multiplier is the engine under the fire sale in stage four of the anatomy — and it compounds with the haircut spiral, because a rising haircut lowers the target leverage at the same time.

The treasurer's early-warning dashboard

Crunches are preceded by measurable stress in funding markets. A treasury that watches the right prices sees the machine spin up before the run. Below are the leading indicators, with rough normal-versus-stressed levels, and how to read each.

Indicator	What it measures	Normal	Stressed (peak)	How to read it
TED spread / FRA-OIS	Unsecured bank funding vs risk-free	~20–30 bp	~450 bp (Oct-08); ~80 bp (Mar-20)	Rising = banks fear lending to each other
Cross-currency basis	Cost of swapping into USD	~0 to −20 bp	−100+ bp	More negative = global dollar shortage
Repo specialness / GC–IORB	Collateral scarcity vs reserve scarcity	GC near IORB	SOFR >> IORB (Sep-19 ~+300bp)	GC above IORB = reserves getting scarce
CP–OIS spread	Short-term corporate funding stress	~10–20 bp	100+ bp	Widening = CP buyers retreating
Off-the-run UST bid-ask	Liquidity of the safest asset	Pennies	Multiples wider (Mar-20)	Wide = even Treasuries illiquid; dash-for-cash
Deposit / redemption velocity	Speed of outflow	<1%/mo	>10%/day (SVB)	Accelerating = run in progress
Collateral haircuts	Lenders' equity demand	Stable	Rising across asset classes	Rising = haircut spiral starting
AT1 / senior CDS	Market's view of bank solvency/funding	Tight, stable	Blows out (CS 2023)	Widening = funding access about to close

No single gauge is decisive; the signal is several of them widening together. The TED/FRA-OIS spread is the classic thermometer of unsecured bank stress; the cross-currency basis reveals dollar shortages that originate offshore; the GC–IORB relationship and repo specialness tell you whether the stress is about scarce reserves (2019) or scarce collateral. CP–OIS and off-the-run Treasury bid-ask widen as the dash-for-cash begins. Deposit/redemption velocity and haircut moves are the on-balance-sheet confirmation that the theory has become your problem. AT1 and CDS spreads are the

market pricing your access to funding — when they blow out, the door is already closing.

> Lend freely, at a high rate, on good collateral. A central bank in a panic must support the market as a whole, not pick which merchant deserves to survive.
>
> — after Walter Bagehot, Lombard Street (1873)

BEST IN CLASS

A world-class bank treasury treats a crunch as a *when*, not an *if*, and engineers survival in advance. It pre-positions collateral at the central bank — assets already pledged and tested at the discount window / SRF so cash is minutes, not days, away. It holds committed, irrevocable facilities rather than relying on the goodwill of overnight markets. It diversifies funding across non-correlated sources — insured retail, term wholesale, secured and unsecured, by currency and geography — so no single door closing is fatal, and it caps concentration in any one depositor or counterparty. It runs daily survival-horizon and reverse-stress tests: not “how bad is a standard stress” but “what exact combination of outflows kills us, and how close are we to it today?” It maintains intraday liquidity visibility — knowing its cash and collateral position in real time, not at end-of-day — because modern runs move in hours. And it keeps a living contingency funding plan with named, pre-agreed triggers: specific indicator levels (FRA-OIS, deposit velocity, haircut moves) that automatically escalate from monitoring to action, with the asset-sale waterfall, the facilities to draw, and the people to call written down before the phone rings. The firms that survived every crunch in the catalogue above had done this work in the calm; the ones that failed were discovering their survival horizon for the first time on the day it reached zero.

12 · BEST IN CLASS — CAPSTONE

What "Best in Class" Looks Like

This capstone draws the whole book together into a target operating model for a world-class bank treasury: one golden source of data, real-time liquidity, an FTP that actually changes behaviour, an integrated funding/XVA desk, collateral run as a managed P&L, strong governance, and the KPI dashboard the best treasurers watch. It closes with a maturity ladder and the canon that makes an individual an expert.

The target operating model

Everything in the preceding chapters — funding, ALM, collateral, capital, XVA, intraday, crises — fails in practice for one recurring reason: the data is fragmented. The single highest-leverage investment a treasury makes is a golden source.

> Golden source — a single, authoritative, real-time data layer that holds every funding trade, collateral position, cash flow, limit and liquidity metric across the firm, reconciled to one version of the truth. Desks, risk, finance and treasury all read from it; nobody keeps a private spreadsheet.

On that foundation sit seven pillars of a world-class treasury:

1. One golden source of funding, collateral and liquidity data — real-time, reconciled, firm-wide.
2. Real-time intraday liquidity and forecasting — BCBS 248 metrics live, with a rolling cash-flow forecast and a buffer sized to the intraday peak.
3. An FTP that changes behaviour — transfer prices granular and timely enough that a desk *feels* the cost of the funding, leverage and liquidity it consumes, and acts on it.
4. An integrated XVA / central-funding desk — one desk pricing and hedging CVA, FVA, MVA, KVA and the funding book, so scarce resources are allocated where they earn most.
5. Collateral optimisation as a managed P&L — cheapest-to-deliver run against a funding benchmark, with its savings measured and owned, not treated as back-office housekeeping.
6. Strong governance — an active ALCO, a clear risk appetite, and a live limit framework with escalation.
7. A KPI dashboard the treasurer watches daily (below), tied to limits and to the risk appetite.

> ALCO (Asset-Liability Committee) — the senior committee that owns the balance-sheet structure: it sets the risk appetite, approves limits (LCR/NSFR buffers, duration, encumbrance, FTP curves), reviews the liquidity position and stress results, and escalates breaches. It is where treasury strategy is governed.

> FTP (Funds Transfer Pricing) — the internal pricing mechanism that charges each business for the funding and liquidity it uses and credits each for the stable funding it provides, moving interest-rate and liquidity risk to a central book. Done well, it makes every lending and trading decision bear its true funding cost.

The best-in-class KPI dashboard

A world-class treasurer watches a tight set of ratios every morning, each with a target buffer above its regulatory floor and a hard limit that triggers escalation.

KPI	What it measures	Best-in-class target
LCR	HQLA vs 30-day stressed net outflow	120–135% (buffer over 100%)
NSFR	Stable funding vs 1-yr required funding	110–120%
Survival horizon	Days the firm funds itself in stress, no market access	> 90 days; aim 180+
Encumbrance ratio	Share of assets pledged / unavailable	< 20–25%, trending flat
FTP coverage	Share of balance sheet fully FTP'd	100% of material books
Intraday peak	Max intraday usage vs available liquidity	Peak < 70% of available
Return on RWA	Risk-adjusted return on capital consumed	> cost of equity (~12–14%)

> Survival horizon — the number of days a firm can meet its obligations in a defined stress scenario using only its own liquidity resources, with no access to new unsecured markets. It is the single most honest liquidity metric because it answers the only question that matters in a run: "how long do we have?"

> Encumbrance ratio — the proportion of a bank's assets that are already pledged (as collateral, in repo, or to covered-bond pools) and therefore unavailable to raise emergency liquidity or to protect unsecured creditors. A rising ratio quietly erodes the firm's last line of defence.

The maturity ladder — laggard to leader

Dimension	Laggard	Follower	Leader
Data	Siloed spreadsheets	Batch warehouse, T+1	Real-time golden source
Liquidity	End-of-day only	Daily LCR reporting	Live intraday + forecast
FTP	Flat cost of funds	Term-matched curve	Full behavioural FTP
Collateral	Manual, bilateral	Tri-party, some optimisation	Optimised, managed P&L
XVA	Priced after the fact	Central CVA desk	Full integrated XVA desk
Governance	Reactive, post-breach	Monthly ALCO	Live limits + risk appetite

A treasury climbs this ladder one dimension at a time, but the order matters: the golden source and real-time liquidity come first, because every other capability — FTP that bites, collateral P&L, integrated XVA, live governance — depends on trustworthy data arriving fast enough to act on. A leader is not the firm with the most dashboards; it is the firm whose treasurer can answer "how much liquidity do we have right now, and for how long?" in seconds, and whose pricing already reflects the answer.

Becoming recognised as an expert — the canon

Technical mastery is necessary but not sufficient; the market recognises experts by their command of a shared canon, vocabulary and set of frameworks. Read the primary sources, not the summaries.

- Stigum's Money Market (Stigum & Crescenzi) — the definitive reference on money markets, repo, and the plumbing of short-term funding. The desk bible.
- Zoltan Pozsar on the *shadow banking system* and market plumbing — the "Money View" of collateral, repo and dealer balance sheets; essential for understanding 2019 and 2020.
- Perry Mehrling, *The New Lombard Street* and the "Money View" lectures — the hierarchy of money and the central bank as dealer of last resort.
- Darrell Duffie on dealer balance sheets, OTC markets, SLR and Treasury-market structure — the academic authority on why intermediation capacity binds.
- Jon Gregory, *The xVA Challenge* — the standard text on counterparty risk, CVA and the full XVA family.
- Primary regulation — the Basel III LCR/NSFR/leverage texts, BCBS 248, the ISDA SIMM methodology, and the ISDA Master/CSA documents themselves. Experts cite the rule, not a blog about it.

> Markets can remain illiquid longer than a leveraged balance sheet can remain solvent; the treasurer's job is to still be standing when liquidity returns.
>
> — a desk maxim, after the 2019–2020 funding shocks

The vocabulary — EPE, PFE, cheapest-to-deliver, SIMM aggregation, SLR headroom, survival horizon, the Method-2 score, the collateral waterfall — is not jargon for its own sake; it is the compression that lets practitioners reason fast about a complex balance sheet. Command the frameworks (the XVA stack, the capital

stack, the default waterfall, the FTP curve, the BCBS 248 tools) and you can place any new instrument, rule or shock into a structure you already understand. That is what separates someone who knows the terms from someone the market trusts to run the book.

BEST IN CLASS

The best-in-class treasurer is defined by a single test: on any morning, from one golden source, they can state the firm's LCR, NSFR, survival horizon, encumbrance, intraday peak and return on RWA; name the binding constraint today; and show that the price the firm quotes already reflects every scarce resource a trade consumes. Behind that sit a real-time data layer, an FTP that changes behaviour, an integrated XVA and funding desk, collateral optimisation run as a managed P&L, and an ALCO with live limits and a clear risk appetite. Everything in this book converges here: a treasury that prices, funds and governs the balance sheet as one connected system — and is still standing, with dry powder, when the next funding shock arrives.

13 · REFERENCE

Glossary & Further Reading

A fast reference to the vocabulary used across this guide, followed by the canon a serious treasury professional is expected to know.

A. Glossary of key terms

Funding & money markets

Maturity transformation — funding long-dated assets with short-dated liabilities; the source of both bank profit and bank fragility.

Repo (repurchase agreement) — a secured loan dressed as a sale and repurchase of securities; the borrower sells collateral and agrees to buy it back at a slightly higher price (the repo rate).

Haircut — the discount applied to collateral value; a 2% haircut means $100 of bonds supports $98 of cash, protecting the lender against price moves.

General collateral (GC) vs special — GC is any bond from an accepted class financed at the market repo rate; a “special” is a specific bond in such demand that lenders accept a below-market rate to borrow it.

SOFR — the Secured Overnight Financing Rate, the US benchmark that replaced LIBOR, derived from overnight Treasury repo.

OIS — an Overnight Index Swap; swapping a fixed rate against compounded overnight rates. The OIS curve is the market's risk-free discount curve.

Commercial paper (CP) — short-term unsecured notes (up to 270 days) issued at a discount by banks and corporates.

TLAC / MREL — Total Loss-Absorbing Capacity / Minimum Requirement for own funds and Eligible Liabilities; the layer of senior debt that can be “bailed in” to recapitalise a failing bank.

AT1 / CoCo — Additional Tier 1 contingent-convertible bonds that convert to equity or are written down when capital falls below a trigger.

Cross-currency basis — the extra cost (or benefit) of raising one currency synthetically via an FX swap versus borrowing it directly; a persistent deviation from covered interest parity.

Liquidity & ALM

LCR (Liquidity Coverage Ratio) — stock of high-quality liquid assets divided by net cash outflows over a 30-day stress; minimum 100%.

NSFR (Net Stable Funding Ratio) — available stable funding divided by required stable funding over a one-year horizon; minimum 100%.

HQLA — High-Quality Liquid Assets: cash, central-bank reserves and government bonds (Level 1), plus capped, haircut amounts of lower-quality assets (Level 2A/2B).

FTP (Funds Transfer Pricing) — the internal rate at which treasury "buys" funds from deposit-gatherers and "sells" funds to lenders, so each business sees the true cost of liquidity.

IRRBB — Interest Rate Risk in the Banking Book; the risk that changing rates erode net interest income (NII) or the economic value of equity (EVE).

EVE vs NII — EVE is the present value of the whole banking book under a rate shock (a value view); NII is the earnings over the next 1–3 years (an income view).

Deposit beta — the fraction of a rate move a bank passes through to depositors; a 50% beta means a +100bp move lifts deposit rates by 50bp.

Non-maturity deposit (NMD) — a deposit with no contractual maturity (checking/savings) whose behavioural life must be modelled.

Collateral, capital & XVA

CSA (Credit Support Annex) — the ISDA document that governs collateral: eligible assets, thresholds, minimum transfer amount and frequency.

Variation margin (VM) — collateral that moves daily to cover mark-to-market gains and losses.

Initial margin (IM) — collateral held against potential future exposure during a close-out; posted to a segregated account and, under UMR, not rehypothecable.

UMR / SIMM — Uncleared Margin Rules require two-way IM above an AANA threshold; ISDA SIMM is the standard risk-based model for computing it.

Rehypothecation — reusing collateral a counterparty posted to you as collateral for your own obligations.

RWA (risk-weighted assets) — assets scaled by riskiness; the denominator of capital ratios.

CET1 ratio — Common Equity Tier 1 capital divided by RWA; the headline solvency measure.

SLR (Supplementary Leverage Ratio) — Tier 1 capital divided by total leverage exposure (on- and off-balance-sheet), irrespective of risk.

G-SIB surcharge — extra capital a Global Systemically Important Bank must hold, scaled to its systemic footprint.

XVA — the family of valuation adjustments to a derivative: CVA (counterparty credit), DVA (own credit), FVA (funding), MVA (initial-margin funding), KVA (capital) and ColVA (collateral).

EPE / PFE — Expected Positive Exposure (the average future exposure) and Potential Future Exposure (a high-percentile exposure); the inputs to CVA and limits.

Survival horizon — how many days a bank can meet its obligations under stress using only its liquidity buffer and contractually certain inflows.

Encumbrance ratio — the share of a bank's assets already pledged as collateral and therefore unavailable in a crisis.

B. Further reading — the canon

- Marcia Stigum & Anthony Crescenzi, *Stigum's Money Market* — the definitive reference on repo, money markets and funding mechanics.
- Zoltan Pozsar (Credit Suisse / ex-Fed research) — essential writing on the market's plumbing, reserves, repo and collateral.
- Perry Mehrling, *The New Lombard Street* — the “money view” and the dealer-of-last-resort framework.
- Darrell Duffie, *How Big Banks Fail* and work on repo runs and dealer balance sheets.
- Jon Gregory, *The xVA Challenge* — the standard practitioner text on counterparty risk and valuation adjustments.
- Basel Committee standards — LCR (BCBS 238), NSFR (BCBS 295), intraday liquidity (BCBS 248), and the Basel III capital and leverage framework.
- Moorad Choudhry, *The Principles of Banking* and *Bank Asset and Liability Management* — practitioner depth on ALM, FTP and liquidity.

DISCLAIMER

This guide is an educational reference. Figures in worked examples are illustrative and chosen to show method, not to quote any specific institution. Regulatory parameters (run-off factors, risk weights, thresholds) are simplified for teaching and evolve across jurisdictions and over time; always check the current rule text before applying them in practice.